

Mixed models extend the predictor $\eta = \mathbf{x}'\boldsymbol{\beta}$ of linear, generalized linear, and categorical regression models by incorporating random effects or coefficients in addition to the non-random or “fixed” effects $\boldsymbol{\beta}$. Therefore, mixed models are sometimes also called random effects models, and have become quite popular for analyzing longitudinal data obtained from repeated observations on individuals or objects in longitudinal studies. A closely related situation is the analysis of clustered data, i.e., when observations are obtained from objects selected by subsampling primary sampling units (clusters or groups of objects) in cross-sectional studies. For example, clusters may be defined by hospitals, schools, or firms, where data from (possibly small) subsamples of patients, students, or clients are collected. Generally, clustering may result from any data generating mechanism that induces a cluster structure. In any case, the data consist of n_i repeated observations

$$(y_{i1}, \dots, y_{ij}, \dots, y_{in_i}, \mathbf{x}_{i1}, \dots, \mathbf{x}_{ij}, \dots, \mathbf{x}_{in_i})$$

of responses and covariates for each individual or cluster $i = 1, \dots, m$. For longitudinal data, y_{ij} and $\mathbf{x}_{ij}$ denote the observed value of the response and the covariate vector, respectively, for individual i at time t_{ij} , $j = 1, \dots, n_i$, while for clustered data these values are observations for subjects or objects j from cluster i . Mixed models allow estimation of individual- or cluster-specific effects, even in the case of relatively small numbers n_i of repeated individual measurements or sizes n_i of subsamples from clusters. The basic idea is to extend the linear predictor $\eta_{ij} = \mathbf{x}_{ij}'\boldsymbol{\beta}$ for observation y_{ij} with fixed population effects $\boldsymbol{\beta}$ to the linear mixed predictor $\eta_{ij} = \mathbf{x}_{ij}'\boldsymbol{\beta} + \mathbf{u}_{ij}'\boldsymbol{\gamma}_i$. Usually $\mathbf{u}_{ij}$ is a subvector of the covariates, and $\boldsymbol{\gamma}_i$ is a vector of individual- or cluster-specific random effects. The assumption of $\boldsymbol{\gamma}_i$ to be fixed effects as in the standard linear or generalized linear model is often impractical since the number of parameters to be estimated becomes quite large relative to the sample size. On the other hand, the random effects distribution implicitly induces certain regularization properties for the cluster parameters $\boldsymbol{\gamma}_i$. An additional advantage of mixed models is that correlations, induced by repeated observations from individuals or clusters, are taken into account during estimation.

In the following, we first describe linear mixed models (LMMs) with (conditionally) Gaussian responses y_{ij} , making the conventional assumption that the random effects are i.i.d. Gaussian variables. We then extend LMMs by allowing correlated Gaussian random effects. This leads to a very broad class of models that are appropriate for analyzing spatial and spatio-temporal data, as well as for Bayesian approaches to non- and semiparametric regression in Chaps. 8 and 9, in particular in Sects. 8.1.9 and 9.6. Statistical inference is described from a frequentist likelihood-oriented as well as a Bayesian perspective.

The second part of chapter extends generalized linear models for non-Gaussian responses to generalized linear mixed models (GLMMs), such as logit or Poisson regression models. Statistical inference for the GLMM is based on similar, but more complicated concepts of the LMM.

7.1 Linear Mixed Models for Longitudinal and Clustered Data

7.1.1 Random Intercept Models

We start with random intercept models which are among the most simple (albeit quite important) mixed models. For notational simplicity we first restrict ourselves to the case of just one covariate x . Let

$$(y_{ij}, x_{ij}), \quad i = 1, \dots, m, \quad j = 1, \dots, n_i$$

denote the values of the response variable y and covariate x observed at times $t_{i1} < \dots < t_{ij} < \dots < t_{in_i}$ for individuals $i = 1, \dots, m$ in a longitudinal study, or for subjects $j = 1, \dots, n_i$ in clusters $i = 1, \dots, m$. Our starting point for modeling the relationship between y and x is the classical linear model

$$y_{ij} = \beta_0 + \beta_1 x_{ij} + \varepsilon_{ij}, \quad (7.1)$$

with i.i.d. errors $\varepsilon_{ij} \sim N(0, \sigma^2)$. In this model, the fact that we have *repeated measurements* $j = 1, \dots, n_i$ on the *same individual or cluster* i is not taken into account. In particular, we not only assume that the observations y_{ij} and y_{rl} , of different individuals i and r , are (stochastically) independent but also repeated measurements y_{ij} and y_{il} on the same individual or cluster i . A possible graphical way to check this independence assumption is to estimate and plot separate regression lines (or more generally regression curves) for each individual or cluster i . If there is no cluster-specific heterogeneity all regression lines should have similar (not identical due to sampling variability) intercepts and slopes. A typical plot is shown in the left panel of Fig. 7.1. The estimates are based on artificial data drawn from the model $y_{ij} = 1 + x_{ij} + \varepsilon_{ij}$, $i = 1, \dots, 10$, $j = 1, \dots, 20$, for $m = 10$ individuals or clusters, $n_i = 20$ repeated measurements in each cluster, and i.i.d. errors $\varepsilon_{ij} \sim N(0, 0.1^2)$. Clearly, the estimated cluster-specific regression lines scatter with low variability

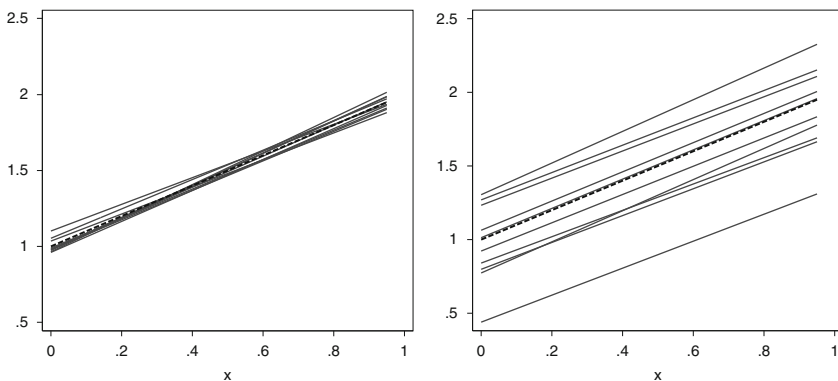


Fig. 7.1 Illustration of random intercept models: both panels show separately estimated regression lines for each cluster. In the *left panel*, there is no cluster-specific random intercept, while in the *right panel* a random intercept is present. The *dashed line* corresponds to the population model

around the true regression line $1 + x$ (dashed line in Fig. 7.1). Hence, there is no reason for assuming cluster-specific heterogeneity and a common regression model for all clusters is sufficient.

The right panel of Fig. 7.1 reveals a different scenario. The estimated cluster-specific regression lines still show a common slope across clusters, but the intercept appears to be different from cluster to cluster. To model this type of cluster-specific heterogeneity we introduce cluster-specific parameters γ_{0i} and obtain

$$y_{ij} = \beta_0 + \beta_1 x_{ij} + \gamma_{0i} + \varepsilon_{ij}, \quad (7.2)$$

where $\varepsilon_{ij} \sim N(0, \sigma^2)$ are the usual i.i.d. errors of the classical linear model. In Eq. (7.2):

- β_0 is the “fixed” population intercept.
- γ_{0i} is the individual- or cluster-specific (random) deviation from the population intercept β_0 .
- $\beta_0 + \gamma_{0i}$ is the (random) intercept for cluster i .
- β_1 is a “fixed” population slope parameter of covariate x that is common across clusters.

Since the individuals or clusters are a random sample from a larger population, the cluster-specific parameters γ_{0i} are assumed to be random with

$$\gamma_{0i} \stackrel{i.i.d.}{\sim} N(0, \tau_0^2). \quad (7.3)$$

We also assume mutual independence between the ε_{ij} and the γ_{0i} . The normal *random effects distribution* in Eq. (7.3) is also sometimes called a *mixture distribution*. The mean can be set to zero because the population mean is already represented by the fixed effect β_0 .

The random intercepts $\beta_0 + \gamma_{0i} \sim N(\beta_0, \tau_0^2)$ may be interpreted as effects of omitted (individual- or cluster-specific) covariates and account for unobserved heterogeneity. Another way to look at the model is to interpret γ_{0i} as an additional error term. The random intercept model appears as a linear regression model with two error terms, where γ_{0i} is then a cluster-level error shared between measurements on the same individual or cluster i and ε_{ij} is the observation error of measurement j in cluster i .

The random intercept model induces a specific correlation or dependence structure on the responses y_{ij} . Given the random intercepts γ_{0i} , the y_{ij} are still conditionally independent with

$$y_{ij} | \gamma_{0i} \sim N(\beta_0 + \beta_1 x_{ij} + \gamma_{0i}, \sigma^2).$$

Marginally, however, repeated measurements y_{ij} for subject or cluster i are correlated with within-subject correlation coefficient

$$\text{Corr}(y_{ij}, y_{il}) = \frac{\tau_0^2}{\tau_0^2 + \sigma^2}, \quad j \neq l; \quad (7.4)$$

see Sect. 7.1.4 for a derivation. Based on the normality assumption of the random effects and the errors, we can further derive the marginal distribution of responses. We have

$$\mathbf{y}_i \sim N(\mathbf{X}_i \boldsymbol{\beta}, \sigma^2 \mathbf{I}_{n_i} + \tau_0^2 \mathbf{J}_{n_i}), \quad (7.5)$$

where $\mathbf{X}_i$ is an $(n_i \times 2)$ -design matrix with ones in the first column and the observed x_{ij} in the second column, and $\mathbf{J}_{n_i}$ denotes an $(n_i \times n_i)$ -matrix of ones. Between two subjects i and r , the observations y_{ij} and y_{rl} are still uncorrelated. The strength of the within-subject correlation Eq. (7.4) depends on the magnitude of the error variances τ_0^2 and σ^2 . The higher the random effects variance τ_0^2 relative to the error variance σ^2 , the stronger the within-subject correlation. Note also that the within-subject correlation is constant (equicorrelation) from measurement to measurement. This may be questionable for longitudinal data, where we might expect correlations dying off for measurements which are farther apart in time.

The marginal model (7.5) also shows what happens with the estimates for $\boldsymbol{\beta}$ if the classical linear model (7.1) is estimated instead of the random intercept model (7.2). If the assumed correlation structure induced by the random intercept model is correct, estimating a classical linear model means that we mistakenly assume an error covariance matrix $\sigma^2 \mathbf{I}$ instead of a non-diagonal covariance matrix as imposed in the marginal model (7.5) of the random intercept model. The consequences of using an incorrect covariance matrix have already been established in Sect. 4.1.1 in the context of the general linear model. As stated there, the estimates for the “fixed” regression coefficients $\boldsymbol{\beta}$ are still unbiased. However, the covariance matrix of $\boldsymbol{\beta}$ and all derived quantities in particular standard errors, confidence intervals, and tests are not correct. Note that the standard errors could either be smaller or larger,

as in a misspecified classical linear model. A nice description of the consequences of mistakenly specifying a classical linear model is also given in [Skrondal and Rabe-Hesketh \(2008\)](#) in Sect. 3.10.1.

Between- and Within-Cluster Effects

The random intercept model (7.2) considered thus far has an important limitation: The so-called within- and between-cluster effects of x are the same. The within-cluster effect denotes the effect if x changes within the *same individual* at different occasions. The between-cluster effect refers to different x values *between different subjects or clusters*. In either case, a difference of one unit of x in model (7.2) induces a difference of β_1 in expected responses. This equality between the within- and between-cluster effect might be questionable in applications. A possible example is our data on undernutrition in Zambia. Here the individual data on children are nested within the districts of Zambia as the cluster variable. Suppose we are interested in the effect of the households wealth, measured by a wealth index, x say, on the Z-score. Now we focus our attention on two imaginary districts, one comparably rich district i and a second rather poor district l . More precisely, we assume that in district i the population is on average richer than in the second district l , i.e., $\bar{x}_i > \bar{x}_l$ with $\bar{x}_i, \bar{x}_l$ being the cluster averages of the wealth index. Suppose first that we compare a child living in the rich district with a child living in the poor district (between district comparison). It is then conceivable that the child living in the rich environment has a higher Z-score, i.e., is better nourished, than the child in the poor environment, even if the individual household wealth is identical. The reason might be that the child profits from the rich environment independent of the individual household situation that might be much less favorable. Economists call this an *external effect*. A possible way to model this between-cluster effect is simply to include the cluster averages $\bar{x}_i$ as an additional covariate into the regression equation. On the other hand, there might be also a within-cluster effect if the individual household wealth is different to that of the average cluster wealth. Children living in households which are wealthier than the average households in the district should have an even higher Z-score (at least on average). This effect may be equal in size to the between-cluster effect but could just as well differ, i.e., being smaller or larger in size. An illustration of different within- and between-cluster effects is shown in Fig. 7.2. The left panel shows different sizes of the within- and between-cluster effects but with identical signs. In the right panel, the signs of the within- and between-cluster effects are opposite of each other.

To deal with possibly different within- and between-cluster effects we can incorporate two covariates derived from x into the predictor: The between-cluster effect can be modeled by the respective cluster means $\bar{x}_i$ as a covariate. The within-cluster effect is modeled by incorporating the individual difference from the cluster mean $x_{ij} - \bar{x}_i$. This yields the extended random intercept model

$$y_{ij} = \beta_0 + \beta_1(x_{ij} - \bar{x}_i) + \beta_2\bar{x}_i + \gamma_{0i} + \varepsilon_{ij}, \quad (7.6)$$

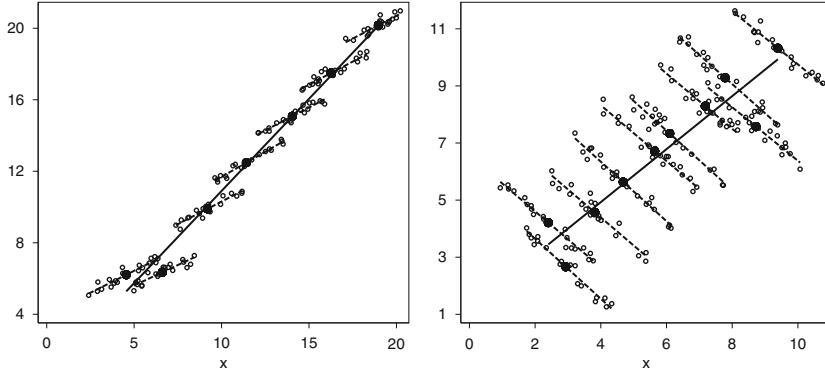


Fig. 7.2 Illustration of within- and between-cluster effects: the between-cluster effects are visualized through the cluster means $\bar{x}_i$ marked by *black dots* and the corresponding linear trends (*solid lines*) which are increasing with x in both panels. The within-cluster effects are illustrated by *dashed lines*

where the coefficient β_2 of the cluster mean represents the between-cluster effect, and the coefficient β_1 of the individual deviation from the cluster mean represents the within-cluster effect. The model collapses to the original random intercept model (7.2) if the within- and between-cluster effects are identical, i.e., if $\beta_1 = \beta_2$. Another interpretation of model (7.6) is obtained by considering $\tilde{\gamma}_{0i} = \beta_2 \bar{x}_i + \gamma_{0i}$ as a random intercept that depends through $\beta_2 \bar{x}_i$ on the covariates.

Alternative Views on the Random Intercept Model

For some applications, the view of the γ_{0i} as random effects can be questionable. This is the case when we do not have the interpretation that clusters are randomly sampled from a larger population. Such a situation arises, for example, when data have been observed on a discrete spatial grid. A typical example is the data for the Munich rent index where the district of each apartment in Munich is given. To account for spatial heterogeneity it might be useful to add a district-specific effect into the predictor; see Example 9.2 of Chap. 9. Since the districts cannot be seen as a random sample of a larger “population” of districts, the interpretation of the district-specific effects as random effects is somewhat artificial. However, there are alternative useful interpretations of the random intercept model which are more suitable for the given situation. In particular, the random effects distribution Eq. (7.3) can be readily understood as the prior for γ_{0i} in a corresponding Bayesian approach. In fact Eq. (7.3) is identical to the Bayesian ridge prior of Sect. 4.4.2. Assuming noninformative priors for the “fixed” effects, i.e., $p(\beta_0) \propto \text{const}$ and $p(\beta_1) \propto \text{const}$, the posterior is given by

$$p(\beta_0, \beta_1, \boldsymbol{\gamma} \mid \mathbf{y}) \propto L(\beta_0, \beta_1, \boldsymbol{\gamma}) \prod_{i=1}^m \frac{1}{\sqrt{\tau_0^2}} \exp\left(-\frac{1}{2\tau_0^2} \gamma_{0i}^2\right),$$

where $\boldsymbol{\gamma} = (\gamma_{01}, \dots, \gamma_{0m})'$ is the vector of cluster effects and $L(\cdot)$ is the Gaussian likelihood of the conditional model. The variances σ^2 and τ_0^2 are assumed fixed for the moment. Now the posterior mode can be obtained by maximizing the log-posterior resulting in the optimization criterion

$$\text{PLS}(\beta_0, \beta_1, \boldsymbol{\gamma}) = \sum_{i=1}^m \sum_{j=1}^{n_i} (y_{ij} - \beta_0 - \beta_1 x_{ij} - \gamma_{0i})^2 + \lambda \sum_{i=1}^m \gamma_{0i}^2 \quad (7.7)$$

with $\lambda = \sigma^2/\tau_0^2$. This has the form of a penalized least squares criterion quite similar to ridge regression outlined in Sect. 4.2.2. To understand the nature of the penalization, we consider the particularly simple random intercept model

$$y_{ij} = \beta_0 + \gamma_{0i} + \varepsilon_{ij}, \quad (7.8)$$

without any covariates, and as usual with $\gamma_{0i} \stackrel{i.i.d.}{\sim} N(0, \tau_0^2)$, $\varepsilon_{ij} \stackrel{i.i.d.}{\sim} N(0, \sigma^2)$. As will be shown in Sect. 7.3.2, the estimator for the γ_{0i} is given by

$$\hat{\gamma}_{0i} = \frac{n_i \tau_0^2}{\sigma^2 + n_i \tau_0^2} \frac{1}{n_i} \sum_{j=1}^{n_i} (y_{ij} - \hat{\beta}_0),$$

where $\hat{\beta}_0 = \bar{y}$, with $\bar{y}$ the overall mean of the responses. The estimator for the cluster mean $\eta_i = \beta_0 + \gamma_{0i}$ is now given by

$$\hat{\eta}_i = \hat{\beta}_0 + \hat{\gamma}_{0i} = \bar{y} + \frac{n_i \tau_0^2}{\sigma^2 + n_i \tau_0^2} \frac{1}{n_i} \sum_{j=1}^{n_i} (y_{ij} - \bar{y}). \quad (7.9)$$

The term $e_i = \frac{1}{n_i} \sum_{j=1}^{n_i} (y_{ij} - \bar{y})$ can be seen as an average residual for individual i , which is a rather natural estimate of γ_{0i} . This is multiplied by the factor

$$\lambda_i = \frac{n_i \tau_0^2}{\sigma^2 + n_i \tau_0^2} < 1,$$

which is sometimes called a shrinkage effect, because the ad hoc estimate e_i for γ_i is shrunken towards the prior mean 0. The larger the n_i , the closer the weight λ_i is to 1 and the smaller the shrinkage. Additional shrinkage is obtained if the error variance σ^2 is large relative to the random effects variance τ_0^2 .

It is instructive to contrast the estimator Eq. (7.9) with two extreme modeling strategies:

- *Full ignorance of groups*: On the one extreme we could fully ignore the groups and estimate the model $y_{ij} = \beta_0 + \varepsilon_{ij}$ with fixed overall intercept β_0 . This is also called the fully pooled model. Of course the least squares estimator for β_0 (which is then identical to the cluster mean η_i) is given by the overall mean $\bar{y}$ of responses, i.e., $\hat{\eta}_i = \hat{\beta}_0 = \bar{y}$.
- *Full distinction of groups*: The other extreme would be to estimate separate models $y_{ij} = \eta_i + \varepsilon_{ij}$ for each cluster i treating the η_i as fixed parameters without random effects distribution in a model without intercept (to omit collinearity problems). This is also known as fully unpooled estimation. Here, the least squares estimators for the η_i are given by the cluster means $\bar{y}_i$, i.e., $\hat{\eta}_i = \bar{y}_i$.

Now the random effects estimator Eq. (7.9) can be seen as a compromise between the two extreme cases. For large n_i or small σ^2 relative to τ_0^2 , the estimator Eq. (7.9) approaches the fully unpooled estimator $\bar{y}_i$. For small n_i or large σ^2 relative to τ_0^2 the fully pooled estimator $\bar{y}$ is approached. In the extreme cases $n_i = 0$ or $\tau_0^2 \rightarrow 0$ or $\sigma^2 \rightarrow \infty$ the mean $\bar{y}$ is reached as a limit. In the other extreme cases $n_i \rightarrow \infty$ or $\tau_0^2 \rightarrow \infty$ or $\sigma^2 \rightarrow 0$ we reach the cluster mean $\bar{y}_i$ as a limit. An illustration of the shrinkage factor λ_i depending on the cluster size n_i (panel a), the random effects variance τ_0^2 (panel b), and the error variance σ^2 (panel c) can be found in Fig. 7.3.

Key Features of Mixed Models

Although the random intercept model is comparably simple it already reveals the key features and advantages of mixed models:

- Individual- or cluster-specific effects can be introduced to account for specific deviations from the population behavior.
- They allow to correct for unobserved heterogeneity induced by omitted covariates.
- Correlations between observations of the same individual or cluster can be taken into account (at least to some extent). This ensures that inference regarding the regression coefficients is correct in the sense that we obtain correct standard errors, confidence intervals, and tests.
- Estimation is stabilized by assuming a common random effects distribution that acts as a penalty term for the otherwise unpenalized cluster-specific effects.

We finally summarize the various interpretations of the random intercept model:

- A *classical interpretation*, where clusters are a random sample of a larger population and the γ_{0i} are cluster-specific random effects.
- A *marginal interpretation*, where the random effects γ_{0i} induce the general linear model (7.5) with correlated errors for the observed y_{ij} .
- A *Bayesian point of view*, which interprets Eq. (7.3) as an underlying prior.
- A *penalized least squares view*, where the penalized least squares criterion induces a penalty on the cluster-specific effects to regularize the estimated parameters.

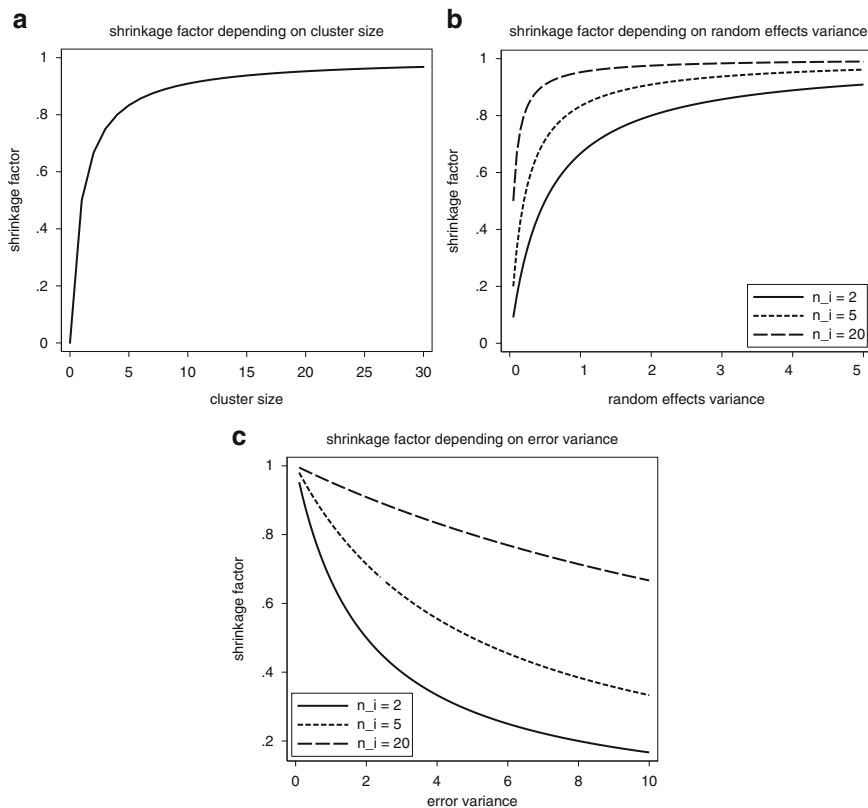


Fig. 7.3 Shrinkage factor in simple random intercept models: Panel (a) illustrates the shrinkage factor λ_i depending on the cluster size n_i (for fixed $\sigma^2 = 1$ and $\tau_0^2 = 1$). Panels (b) and (c) display the shrinkage factor depending on the random effects variance τ_0^2 (for fixed $\sigma^2 = 1$) and the error variance σ^2 (for fixed $\tau_0^2 = 1$), respectively

7.1.2 Random Coefficient or Slope Models

Random intercept models are still based on the assumption that the covariate effect of x is equal in size for each individual or cluster. However, this may be too restrictive as Fig. 7.4 reveals. The left panel shows the separately estimated cluster-specific regression lines for some artificial data. Here, the intercept for the regression lines seems to be identical (or at least close), whereas the slopes are clearly different. More typical is the situation in the right panel of Fig. 7.4, where the regression lines exhibit both cluster-specific intercepts and slopes. To cope with such cluster-specific slopes, we extend the random intercept model (7.2) to obtain

$$y_{ij} = \beta_0 + \beta_1 x_{ij} + \gamma_{0i} + \gamma_{1i} x_{ij} + \varepsilon_{ij}, \quad (7.10)$$

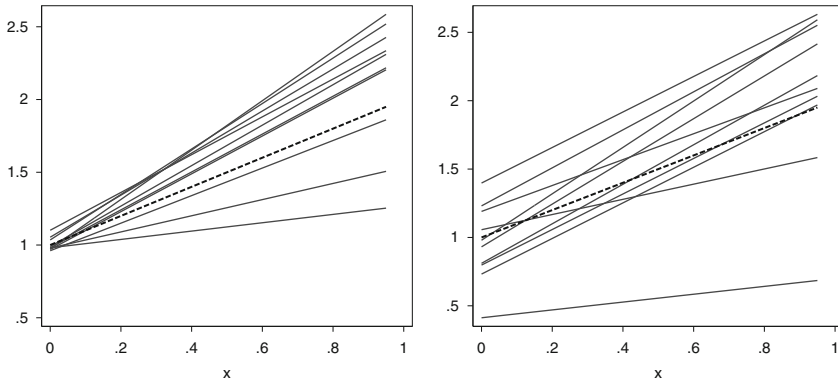


Fig. 7.4 Illustration of random slope models: both panels show separately estimated regression lines for each cluster. In the *left panel*, there is a cluster-specific random slope and a common intercept across clusters. In the *right panel*, both the intercept and the slope are cluster-specific. The *dashed line* corresponds to the population model

where:

- β_1 is the “fixed” population slope of the effect of x .
- γ_{1i} is the individual- or cluster-specific deviation for the slope.
- $\beta_1 x_{ij}$ is the population effect of x .
- $\beta_1 x_{ij} + \gamma_{1i} x_{ij}$ is the cluster-specific effect of x .

The left panel of Fig. 7.4 corresponds to a model with most random intercept parameters γ_{0i} zero or close to zero and random slope parameters γ_{1i} different from zero. In the right panel both random coefficients are mostly found to be away from zero.

For the cluster-specific parameters, we now define the bivariate normal random effects distribution

$$\boldsymbol{\gamma}_i = \begin{pmatrix} \gamma_{0i} \\ \gamma_{1i} \end{pmatrix} \stackrel{i.i.d.}{\sim} N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{10} & \tau_1^2 \end{pmatrix} \right) \quad (7.11)$$

with mean $\mathbf{0} = (0, 0)'$ and covariance matrix

$$\boldsymbol{Q} = \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{10} & \tau_1^2 \end{pmatrix}.$$

The parameters τ_0^2 and τ_1^2 determine the variability of the cluster-specific intercepts and slopes, respectively. The left panel of Fig. 7.4 corresponds to $\tau_0^2 \approx 0$ and $\tau_1^2 > 0$, while in the right panel, $\tau_0^2 > 0$ also holds.

The covariance $\tau_{01} = \tau_{10}$ can capture correlations between random intercepts and slopes. Such a correlation may be present, for example, when individuals with larger slopes tend to have smaller intercepts, leading to negatively correlated random intercepts and slopes.

Similar to simple random intercepts, the random coefficient model (7.10) induces a certain (marginal) correlation structure on observations y_{ij} and y_{ir} from the same cluster. This correlation structure is, however, more complicated as it depends on the covariate values x_{ij} . First of all, the marginal variances of y_{ij} are heteroscedastic with

$$\text{Var}(y_{ij}) = \tau_0^2 + 2\tau_{01}x_{ij} + \tau_1^2 x_{ij}^2 + \sigma^2;$$

see Sect. 7.1.4 for a derivation. Thus, the marginal variances depend quadratically on the covariate values. The covariance between y_{ij} and y_{ir} can be shown to be

$$\text{Cov}(y_{ij}, y_{ir}) = \tau_0^2 + \tau_{01}x_{ij} + \tau_{01}x_{ir} + \tau_1^2 x_{ij}x_{ir}$$

resulting in an intraclass correlation coefficient

$$\text{Corr}(y_{ij}, y_{ir}) = \frac{\text{Cov}(y_{ij}, y_{ir})}{\text{Var}(y_{ij})^{1/2} \text{Var}(y_{ir})^{1/2}},$$

which depends in a rather complicated way on the observed covariate values and is very difficult to interpret.

We now illustrate the idea of random intercept and slope models with a first example, thereby extending our framework in a natural way to more than one covariate. We reconsider Example 2.9 (p. 39). Another application with cluster data and several random slopes can be found in Sect. 7.7, which presents a case study on sales of orange juice.

Example 7.1 Hormone Therapy with Rats—Linear Mixed Model

To investigate the effect of testosterone on their craniofacial growth, 50 male rats have been randomized to either a control group or one of two treatment groups. Treatment consisted of a low or high dose of the drug Decapeptyl, an inhibitor for testosterone production, and started at the age of 45 days. To measure the growth of the skull of each rat, X-ray pictures were taken every 10 days, with the first observation at the age of 50 days. In this example, height of the skull, defined as the distance (in pixels) between two well-defined points, is considered as the response variable. The individual profiles $\{y_{ij}, j = 1, \dots, n_i\}$ for the three groups are shown in Fig. 2.10 (p. 40). Since many rats drop out before the end of the study, the numbers n_i of individual repeated measurements are different; see Table 2.3 (p. 39).

In Example 2.9 (p. 39), we formulated the LMM

$$y_{ij} = \beta_0 + \beta_1 L_i t_{ij} + \beta_2 H_i t_{ij} + \beta_3 C_i t_{ij} + \gamma_{0i} + \gamma_{1i} t_{ij} + \varepsilon_{ij}, \quad (7.12)$$

with the transformed age

$$t = \log(1 + (\text{age} - 45)/10)$$

as the time scale (with $t = 0$ corresponding to begin of treatment), the time points

$$t_{ij} = \log(1 + (\text{age}_{ij} - 45)/10)$$

for measurement j of rat i , and the indicator variables

$$L_i = \begin{cases} 1, & \text{rat } i \text{ in low-dose group,} \\ 0, & \text{otherwise,} \end{cases}$$

$$H_i = \begin{cases} 1, & \text{rat } i \text{ in high-dose group,} \\ 0, & \text{otherwise,} \end{cases}$$

$$C_i = \begin{cases} 1, & \text{rat } i \text{ in control group,} \\ 0, & \text{otherwise.} \end{cases}$$

In this model, β_0 is the population intercept of rats before treatment, γ_{0i} are individual-specific deviations from this population intercept, the effects β_1 , β_2 , and β_3 are the (different) population slopes for the three groups, and γ_{1i} are individual-specific deviations from these slopes. Therefore, $\beta_1 + \gamma_{1i}$, $\beta_2 + \gamma_{1i}$, and $\beta_3 + \gamma_{1i}$ are the individual slopes for rat i , in the low-dose, high-dose, and control group, respectively. Similarly, $\beta_0 + \beta_1 t_{ij} + \gamma_{0i} + \gamma_{1i} t_{ij}$, $\beta_0 + \beta_2 t_{ij} + \gamma_{0i} + \gamma_{1i} t_{ij}$, and $\beta_0 + \beta_3 t_{ij} + \gamma_{0i} + \gamma_{1i} t_{ij}$ are the corresponding linear trends.

In contrast to the “fixed” effects $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2, \beta_3)'$, the individual-specific effects $\boldsymbol{\gamma}_i = (\gamma_{0i}, \gamma_{1i})'$ are considered to be “random” because the rats are randomly selected from a population. We make the usual assumption that the random effects are independent and identically distributed Gaussian random variables according to Eq. (7.11) but with the additional restriction $\tau_{01} = \tau_{10} = 0$.

The variances τ_0^2 and τ_1^2 characterize the amount of variability of individual-specific deviations γ_{0i} from the population intercept β_0 and of deviations γ_{1i} from population slopes, respectively. Note that we implicitly assume that deviations from the three population slopes have the same variance.

A closer look at Fig. 2.10 (p. 40) suggests that individual-specific slopes do not differ much from each other, corresponding to the assumption $\gamma_{1i} = 0$ for all i . This leads to the simplified or reduced model

$$y_{ij} = \beta_0 + \beta_1 L_i t_{ij} + \beta_2 H_i t_{ij} + \beta_3 C_i t_{ij} + \gamma_{0i} + \varepsilon_{ij}, \quad (7.13)$$

where only i.i.d. random intercepts $\gamma_{0i} \sim N(0, \tau_0^2)$ are incorporated, without random slopes.

Table 7.1 contains estimates for the fixed effects and the variance parameters in both models. Estimates are based on methodology outlined in Sect. 7.3 and estimation has been carried out using function `lmer` of the R package `lme4`. Furthermore, Table 7.1 shows the estimated effects of a linear regression model without any random effects.

We first take a look at the random intercept model (7.13). The estimated random effects variance of 3.565 is quite large compared to the overall error variance of 1.445 yielding a comparably high intraclass correlation coefficient of

$$\widehat{\text{Corr}}(y_{ij}, y_{ir}) = \frac{3.565}{3.565 + 1.445} = 0.71.$$

This indicates strong individual-specific heterogeneity. Compared to the linear model without random effects, the estimated population intercepts are in close agreement, whereas estimated treatment effects differ. In particular, the standard errors are higher in the simple linear model, which is in part due to the considerably larger estimate for the error variance σ^2 that has “captured” omitted individual-specific random effects. Comparing the random intercept model (7.13) with the more complex random slope model (7.12), all estimates are

Table 7.1 Hormone therapy with rats: estimates including standard errors for the random slope model, the reduced random intercept model, and a classical linear model without any random effects

	Parameter	Model (7.12)		Model (7.13)		Linear model	
		Estimate	s.e.	Estimate	s.e.	Estimate	s.e.
<i>intercept</i>	β_0	68.606	0.327	68.607	0.331	68.687	0.348
<i>low-dose</i>	β_1	7.505	0.227	7.507	0.225	7.677	0.286
<i>high-dose</i>	β_2	6.875	0.230	6.871	0.228	6.529	0.284
<i>control</i>	β_3	7.318	0.284	7.314	0.281	7.212	0.326
$\text{Var}(\gamma_{0i})$	τ_0^2	3.430		3.565			
$\text{Var}(\gamma_{1i})$	τ_1^2	0.001					
$\text{Var}(\varepsilon_{ij})$	σ^2	1.444		1.445		4.730	

very close. Moreover, the variance for the random slope is very small (0.001), so that the simpler random intercept model may be sufficient in this application. The decision as to whether or not the reduced model (7.13) is appropriate can be also supported by a statistical test; see Sect. 7.3.4.

△

7.1.3 General Model Definition and Matrix Notation

We now define LMMs in the more general matrix notation containing the illustrative models and examples of the previous sections as special cases. Let

$$(y_{ij}, \mathbf{x}'_{ij}), \quad i = 1, \dots, m, \quad j = 1, \dots, n_i,$$

denote the values of a response variable y and a covariate vector $\mathbf{x}$ observed at times $t_{i1} < \dots < t_{ij} < \dots < t_{in_i}$ for individuals $i = 1, \dots, m$ in a longitudinal study (as in Example 7.1) or for subjects $j = 1, \dots, n_i$ in cluster $i = 1, \dots, m$.

A LMM is hierarchically defined through the following stages. In the *first stage*, responses y_{ij} are assumed to depend linearly on unknown fixed population effects $\boldsymbol{\beta}$ and on unknown individual- or cluster-specific effects $\boldsymbol{\gamma}_i$ through the *measurement model*

$$y_{ij} = \mathbf{x}'_{ij}\boldsymbol{\beta} + \mathbf{u}'_{ij}\boldsymbol{\gamma}_i + \varepsilon_{ij}.$$

In this case, $\mathbf{x}_{ij} = (1, x_{ij1}, \dots, x_{ijk})'$, $\mathbf{u}_{ij} = (1, u_{ij1}, \dots, u_{ijq})'$, and ε_{ij} represent design vectors and the error terms, respectively; the latter are assumed to be i.i.d. $N(0, \sigma^2)$ random variables. The design vectors include original covariates from the given data set or can be constructed from original covariates. The components may be time-dependent, such as the transformed age t in Example 7.1, or may be time-constant. For clustered data, this corresponds to covariates that vary across subjects or only across clusters. Usually, $\mathbf{x}_{ij}$ as well as $\mathbf{u}_{ij}$ both contain the constant 1, so that the linear predictor includes a fixed population intercept β_0 and

a random intercept γ_{0i} . In addition, $\mathbf{u}_{ij}$ is typically a subvector of $\mathbf{x}_{ij}$ and will not contain additional components. As an example consider the simple random intercept model (7.2), where $\mathbf{x}_{ij} = (1, x_{ij})'$ and $\mathbf{u}_{ij} = 1$. In the extended random intercept model (7.6), we have $\mathbf{x}_{ij} = (1, x_{ij} - \bar{x}_i, \bar{x}_i)'$ and $\mathbf{u}_{ij} = 1$, while for the simple random slope model (7.10) we obtain $\mathbf{x}_{ij} = (1, x_{ij})'$ and $\mathbf{u}_{ij} = (1, x_{ij})'$.

Collecting all individual- or cluster-specific responses y_{ij} , design vectors $\mathbf{x}_{ij}$, $\mathbf{u}_{ij}$, and errors ε_{ij} , $j = 1, \dots, n_i$, into vectors or design matrices

$$\mathbf{y}_i = \begin{pmatrix} y_{i1} \\ \vdots \\ y_{ij} \\ \vdots \\ y_{in_i} \end{pmatrix}, \quad \mathbf{X}_i = \begin{pmatrix} \mathbf{x}'_{i1} \\ \vdots \\ \mathbf{x}'_{ij} \\ \vdots \\ \mathbf{x}'_{in_i} \end{pmatrix}, \quad \mathbf{U}_i = \begin{pmatrix} \mathbf{u}'_{i1} \\ \vdots \\ \mathbf{u}'_{ij} \\ \vdots \\ \mathbf{u}'_{in_i} \end{pmatrix}, \quad \boldsymbol{\varepsilon}_i = \begin{pmatrix} \varepsilon_{i1} \\ \vdots \\ \varepsilon_{ij} \\ \vdots \\ \varepsilon_{in_i} \end{pmatrix}, \quad (7.14)$$

we obtain the measurement model in matrix notation

$$\mathbf{y}_i = \mathbf{X}_i \boldsymbol{\beta} + \mathbf{U}_i \boldsymbol{\gamma}_i + \boldsymbol{\varepsilon}_i \quad (7.15)$$

for individual or cluster $i = 1, \dots, m$ with $E(\boldsymbol{\varepsilon}_i) = \mathbf{0}$.

In the *second stage*, the model is supplemented through a distributional assumption for the random effects $\boldsymbol{\gamma}_i$, reflecting the idea that the data are generated by drawing individuals or clusters from a population. We assume that the random effects are independent and identically distributed according to a *random effects* or—from a Bayesian perspective—a *prior* distribution. In this book, we make the conventional assumption of a Gaussian random effects distribution

$$\boldsymbol{\gamma}_i \sim N(\mathbf{0}, \mathbf{Q}),$$

with unknown $(q+1) \times (q+1)$ -covariance matrix $\mathbf{Q}$. The covariance matrix may be of a specific structural form. In particular, the special case of (a priori) independent components γ_{il} of $\boldsymbol{\gamma}_i$ corresponds to a diagonal matrix $\mathbf{Q} = \text{diag}(\tau_0^2, \dots, \tau_q^2)$ with elements $\tau_l^2 = \text{Var}(\gamma_{il})$, $l = 0, \dots, q$. Such models are sometimes called *variance components models* and are among the least complex random effects models.

The assumption of i.i.d. $N(0, \sigma^2)$ errors implies

$$\boldsymbol{\varepsilon}_i \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_{n_i}), \quad (7.16)$$

as well as independence of error vectors $\boldsymbol{\varepsilon}_1, \dots, \boldsymbol{\varepsilon}_m$. Additionally, error terms and random effects are assumed to be mutually independent. Correlation between repeated observations within individuals or clusters is, however, induced by the common vector $\boldsymbol{\gamma}_i$ of random effects, as is illustrated for simple random intercept and slope models in the previous Sects. 7.1.1 and 7.1.2.

Additional correlation can be specified through

$$\boldsymbol{\varepsilon}_i \sim \mathbf{N}(\mathbf{0}, \boldsymbol{\Sigma}_i)$$

with a non-diagonal $n_i \times n_i$ -covariance matrix $\boldsymbol{\Sigma}_i$. Special forms for $\boldsymbol{\Sigma}_i$ result from assumptions about the error process $\{\varepsilon_{ij}\}$. For example, the assumption of autoregressive errors leads to covariance matrices as considered in Sect. 4.1.4. If the errors follow autoregressive processes of first order, i.e.,

$$\varepsilon_{ij} = \rho \varepsilon_{i,j-1} + \tilde{\varepsilon}_{ij}, \quad \tilde{\varepsilon}_{ij} \sim \mathbf{N}(0, \sigma_{\varepsilon}^2), \quad 0 < \rho < 1,$$

and if observations are made at the same equidistant time points $t_1 < t_2 < \dots < t_n$, then all covariance matrices $\boldsymbol{\Sigma}_i$ have the form (4.5) (see p. 192). More generally, the correlations $\rho^{|j-j'|}$ in Eq. (4.5) have to be replaced by $\rho^{|t_{ij}-t_{ij'}|}$, $j, j' = 1, \dots, n_i$. In our examples, we will usually adhere to the stronger assumption (7.16) of i.i.d. errors. A summary of LMMs for longitudinal and clustered data can be found in Box 7.1.

The following example illustrates how the LMM Eq. (7.12) for the rats data is reexpressed in the general form (7.14) and (7.15).

Example 7.2 Hormone Therapy with Rats—Matrix Notation

Measurements are available for rats at the age of $a = 50, 60, \dots, 110$ days, as long as they remain in the study. After the logarithmic transformation $t = \log(1 + (\text{age} - 45)/10)$, time corresponds to the values

$$t_1 = \log(1.5), \quad t_2 = \log(2.5), \quad \dots, \quad t_7 = \log(7.5).$$

For rat 17 from the low-dose group ($L_{17} = 1, H_{17} = 0, C_{17} = 0$), $n_{17} = 5$ measurements have been made. This results in

$$\mathbf{y}_{17} = \begin{pmatrix} y_{17,1} \\ \vdots \\ y_{17,5} \end{pmatrix}, \quad \mathbf{X}_{17} = \begin{pmatrix} 1 & \log(1.5) & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & \log(5.5) & 0 & 0 \end{pmatrix}, \quad \mathbf{U}_{17} = \begin{pmatrix} 1 & \log(1.5) \\ \vdots & \vdots \\ 1 & \log(5.5) \end{pmatrix}$$

for model (7.12). For rat 12 from the high-dose group ($L_{12} = 0, H_{12} = 1, C_{12} = 0$) with $n_{12} = 4$ measurements, we obtain

$$\mathbf{y}_{12} = \begin{pmatrix} y_{12,1} \\ \vdots \\ y_{12,4} \end{pmatrix}, \quad \mathbf{X}_{12} = \begin{pmatrix} 1 & 0 & \log(1.5) & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & 0 & \log(4.5) & 0 \end{pmatrix}, \quad \mathbf{U}_{12} = \begin{pmatrix} 1 & \log(1.5) \\ \vdots & \vdots \\ 1 & \log(4.5) \end{pmatrix},$$

and for rat 22 from the control group ($L_{22} = 0, H_{22} = 0, C_{22} = 1$),

$$\mathbf{y}_{22} = \begin{pmatrix} y_{22,1} \\ \vdots \\ y_{22,7} \end{pmatrix}, \quad \mathbf{X}_{22} = \begin{pmatrix} 1 & 0 & 0 & \log(1.5) \\ \vdots & \vdots & \vdots & \vdots \\ 1 & 0 & 0 & \log(7.5) \end{pmatrix}, \quad \mathbf{U}_{22} = \begin{pmatrix} 1 & \log(1.5) \\ \vdots & \vdots \\ 1 & \log(7.5) \end{pmatrix}.$$

7.1 Linear Mixed Model for Longitudinal and Clustered Data

Measurement Model

LMM for longitudinal or clustered data is given by

$$y_{ij} = \mathbf{x}'_{ij}\boldsymbol{\beta} + \mathbf{u}'_{ij}\boldsymbol{\gamma}_i + \varepsilon_{ij}$$

for individuals or clusters $i = 1, \dots, m$ observed at occasions

$$t_{i1} < \dots < t_{ij} < \dots < t_{in_i},$$

or in matrix notation

$$\mathbf{y}_i = \mathbf{X}_i\boldsymbol{\beta} + \mathbf{U}_i\boldsymbol{\gamma}_i + \boldsymbol{\varepsilon}_i.$$

In this model $\mathbf{y}_i$ is the n_i -dimensional vector of responses for individual or cluster i , $\mathbf{X}_i$ and $\mathbf{U}_i$ are $n_i \times p$ - and $n_i \times (q+1)$ -dimensional design matrices constructed from known covariates, $\boldsymbol{\beta}$ is the p -dimensional vector of fixed effects, $\boldsymbol{\gamma}_i$ is a $(q+1)$ -dimensional vector of individual- or cluster-specific effects, and $\boldsymbol{\varepsilon}_i$ is a n_i -dimensional vector of errors.

Distributional Assumptions

For $\boldsymbol{\gamma}_i$ and $\boldsymbol{\varepsilon}_i$, $i = 1, \dots, m$, the following distributional assumptions hold:

$$\boldsymbol{\gamma}_i \sim N(\mathbf{0}, \boldsymbol{Q}), \quad \boldsymbol{\varepsilon}_i \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_{n_i}).$$

$\boldsymbol{\gamma}_1, \dots, \boldsymbol{\gamma}_m, \boldsymbol{\varepsilon}_1, \dots, \boldsymbol{\varepsilon}_m$ are assumed to be independent.

Software

- Functions `xtreg`, `xtmixed`, and `gllamm` of STATA; see [Skron dal and Rabe-Hesketh \(2008\)](#)
- Package `lme4` of R
- `proc mixed` of SAS
- Software package BayesX; see also the R interface `R2BayesX`

For model (7.13) without random slopes $\gamma_{li} \cdot t_{ij}$, the second column needs to be deleted in the design matrices $\mathbf{U}_{12}$, $\mathbf{U}_{17}$, and $\mathbf{U}_{22}$.

△

7.1.4 Conditional and Marginal Formulation

The measurement model (7.15) implies the *conditional Gaussian model*

$$\mathbf{y}_i | \boldsymbol{\gamma}_i \sim N(\mathbf{X}_i \boldsymbol{\beta} + \mathbf{U}_i \boldsymbol{\gamma}_i, \sigma^2 \mathbf{I})$$

for the response vector $\mathbf{y}_i$, given the random effect $\boldsymbol{\gamma}_i$. From this conditional perspective, the individual- or cluster-specific effects $\boldsymbol{\gamma}_i$ are interpreted similarly as the usual regression effects, with the difference that they only apply to individual or cluster i . Rewriting the model (7.15) in the form

$$\mathbf{y}_i = \mathbf{X}_i \boldsymbol{\beta} + \boldsymbol{\varepsilon}_i^*,$$

with errors $\boldsymbol{\varepsilon}_i^* = \mathbf{U}_i \boldsymbol{\gamma}_i + \boldsymbol{\varepsilon}_i$, results in a linear Gaussian regression model with correlated errors

$$\boldsymbol{\varepsilon}_i^* \sim N(\mathbf{0}, \mathbf{V}_i), \quad \mathbf{V}_i = \text{Cov}(\boldsymbol{\varepsilon}_i) + \text{Cov}(\mathbf{U}_i \boldsymbol{\gamma}_i) = \sigma^2 \mathbf{I} + \mathbf{U}_i \mathbf{Q} \mathbf{U}_i'.$$

The first equality for $\mathbf{V}_i$ holds because $\boldsymbol{\varepsilon}_i$ and $\boldsymbol{\gamma}_i$ are assumed to be independent, and $\text{Cov}(\mathbf{U}_i \boldsymbol{\gamma}_i) = \mathbf{U}_i \mathbf{Q} \mathbf{U}_i'$ follows from Theorem B.2, property 5, in Appendix B.2.

Therefore the corresponding *marginal Gaussian model* for $\mathbf{y}_i$ is given by

$$\mathbf{y}_i \sim N(\mathbf{X}_i \boldsymbol{\beta}, \sigma^2 \mathbf{I} + \mathbf{U}_i \mathbf{Q} \mathbf{U}_i').$$

This shows that interpretation of population parameters $\boldsymbol{\beta}$ in the conditional model is the same as in the marginal model. This useful property of LMMs is lost for GLMMs that follow in Sect. 7.5. The non-diagonal covariance matrix $\mathbf{V}_i$ induces a correlation structure between repeated measurements of individual or cluster i in the marginal model, even if the errors $\boldsymbol{\varepsilon}_{ij}$ in the conditional model are uncorrelated. For the special case of a random intercept model, where $\mathbf{U}_i = (1, \dots, 1)'$ and $\mathbf{Q} = \tau_0^2$, we obtain

$$\mathbf{U}_i \mathbf{Q} \mathbf{U}_i' = \tau_0^2 \begin{pmatrix} 1 & \dots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \dots & 1 \end{pmatrix} = \tau_0^2 \mathbf{J}_{n_i},$$

where $\mathbf{J}_{n_i}$ is an $(n_i \times n_i)$ -matrix of ones. The covariance matrix $\mathbf{V}_i$ reduces to the *equi-covariance matrix* of the marginal form of the random intercept model, which has diagonal elements $\text{Var}(\varepsilon_{ij}^*) = \sigma^2 + \tau_0^2$ and off-diagonal elements $\text{Cov}(\varepsilon_{ij}^*, \varepsilon_{il}^*) = \text{Var}(\gamma_i) = \tau_0^2$, $j \neq l$, implying equal correlations

$$\text{Corr}(\varepsilon_{ij}^*, \varepsilon_{il}^*) = \frac{\tau_0^2}{\sigma^2 + \tau_0^2}, \quad j \neq l,$$

for errors in cluster i . Errors from different clusters are assumed to be uncorrelated.

7.1.5 Stochastic Covariates

As in linear and generalized linear models, we have to distinguish between deterministic and stochastic covariates. In designed experiments, as in Example 7.1, covariates are typically deterministic. On the other hand, observational studies have mostly stochastic covariates, i.e., both responses y_i and covariates x_{ij} , $i = 1, \dots, m$, $j = 1, \dots, n_i$, have to be considered as realizations of a random vector (y, x') . Our assumptions on errors and random effects in Box 7.1 are then to be understood conditionally on the covariates, as, for example,

$$\boldsymbol{\varepsilon}_i \mid X_i \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_{n_i}), \quad \boldsymbol{\gamma}_i \mid X_i \sim N(\mathbf{0}, \mathbf{Q}).$$

Since the conditional distributions of $\boldsymbol{\varepsilon}_i$ and $\boldsymbol{\gamma}_i$ do not depend on X_i , these two assumptions imply that the errors and random effects are independent from covariates. For errors, this is completely analogous to linear models; see Sect. 3.1. For random effects, this assumption is likely to be violated if we assume that the random effects are surrogates for omitted covariates, as the omitted covariates may well be correlated with the observed covariates. In the econometrics literature, covariates that are correlated with the errors are called endogenous.

We illustrate the problem of correlation between random effects and observed covariates with the simple random intercept model. Following Sect. 3.2.1 of [Skrondal and Rabe-Hesketh \(2004\)](#), we discuss the situation for the case when the correct model is

$$y_{ij} = \beta_0 + \beta_1 x_{ij} + \beta_2 w_i + \gamma_{0i} + \varepsilon_{ij},$$

and the misspecified working model is

$$y_{ij} = \beta_0 + \beta_1 x_{ij} + \tilde{\gamma}_{0i} + \varepsilon_{ij},$$

where the cluster-specific covariate w_i is omitted. Therefore, $\tilde{\gamma}_{0i} = \beta_2 w_i + \gamma_{0i}$. Assume now that w_i and x_{ij} are dependent in form of the regression model

$$w_i = \alpha_0 + \alpha_1 \bar{x}_i + u_i,$$

where $\bar{x}_i$ is the mean of the x_{ij} , $j = 1, \dots, n_i$. Inserting w_i in the equation for $\tilde{\gamma}_{0i}$ gives

$$\tilde{\gamma}_{0i} = \beta_2 \alpha_0 + \beta_2 \alpha_1 \bar{x}_i + \beta_2 u_i + \gamma_{0i} = \delta_0 + \delta_1 \bar{x}_i + \delta_{0i}$$

after reparameterizing with $\delta_0 = \beta_2 \alpha_0$, $\delta_1 = \beta_2 \alpha_1$, and $\delta_{0i} = \beta_2 u_i + \gamma_{0i}$. Substituting these results into the misspecified model finally yields

$$y_{ij} = \beta_0 + \delta_0 + \delta_1 \bar{x}_i + \beta_1 x_{ij} + \delta_{0i} + \varepsilon_{ij}, \quad (7.17)$$

with the new fixed intercept $\beta_0 + \delta_0$ and the new random intercept δ_{0i} . Therefore, inclusion of $\bar{x}_i$ as a separate covariate retains β_1 as the required effect of x_{ij} , although

the covariate w_i is missing. Omitting $\bar{x}_i$ from the model will, however, lead to a biased estimate of β_1 if $\delta_1 = \beta_2\alpha_1 \neq 0$; see also Sect. 3.4 for a discussion of the effect of missing covariates on the bias in regression models. For interpretational purposes it is often better to include the centered covariate $x_{ij} - \bar{x}_i$ rather than x_{ij} into the model. We can then estimate the extended random intercept model (7.6) discussed in Sect. 7.1.1 and interpret the effects of $\bar{x}_i$ and $x_{ij} - \bar{x}_i$ as between- and within-cluster effects, respectively. For models with several random effects, in particular random slopes, analogous results are available; see [Snijders and Berkhof \(2004\)](#).

In the econometrics literature often another approach to cope with the problem of correlation between random effects and observed covariates is preferred. Here, a fixed effects approach is proposed assuming fixed cluster-specific effects γ_{0i} without assuming a random effects distribution. Of course, inclusion of all γ_{0i} , $i = 1, \dots, m$, into the regression equation is not feasible as then the design matrix is rank deficient. As a remedy we could either omit one of the γ_{0i} 's (the corresponding cluster then serves as the reference category) or omit the global intercept β_0 . Omitting the global intercept we arrive at the model

$$y_{ij} = \beta_1 x_{ij} + \gamma_{0i} + \varepsilon_{ij}, \quad (7.18)$$

where the γ_{0i} are treated as fixed parameters and the cluster index has the role of a categorical covariate albeit with a large number of categories. The advantage of such a specification is that we eliminated the correlation of x_{ij} with the random effect simply with the replacement of fixed effects. Since Eq. (7.18) is a classical linear model both the regression parameter β_1 and the cluster dummies γ_{0i} can be estimated unbiasedly using ordinary least squares. However, since the cluster dummies are not regularized as in the random effects approach, they are usually estimated with considerable variance at least if the cluster sizes n_i are small (as is often the case). Moreover, the fixed effects approach uses only within-cluster information. For covariates with equal between- and within-cluster effect, a random effects approach allows more precise estimation of the effect as also the between-cluster variability is used. Finally, covariates at the cluster level are not possible in a fixed effects approach because of perfect linear dependence between the columns in the design matrix corresponding to the cluster dummies and the cluster-specific variables. Nevertheless, the approach can be useful if interest lies solely in estimating the regression coefficient β_1 , while the cluster dummies are nuisance parameters. It turns out that the regression coefficient β_1 can be estimated without resorting to the γ_{0i} 's. It can be shown that the least squares estimator for β_1 can be obtained by regressing the centered responses $y_{ij} - \bar{y}_i$ on the centered covariate values $x_{ij} - \bar{x}_i$ in a model without intercept; see p. 410 in Sect. 7.8 for a derivation. This result also carries over to more than one covariate, then all covariates need to be centered by their respective cluster means.

In summary, there are two ways to deal with the problem of random intercepts that are correlated with the available covariates. If interest primarily lies in the

fixed effect regression coefficients, we can simply regress the cluster-wise centered responses on the cluster-wise centered covariate values. If we are also interested in estimates for the random effects γ_{0i} and their variance τ_0^2 , a random effects approach with the cluster means of the covariates and the deviations from the means as regressors is preferable. This approach also allows to estimate the between- and within-cluster effects.

7.2 General Linear Mixed Models

The LMM $y_i = X_i\beta + U_i\gamma + \varepsilon_i$, $i = 1, \dots, m$, as well as multilevel extensions and further types of mixed models considered below and in subsequent chapters, can be formulated in a rather compact form by defining the vectors

$$y = \begin{pmatrix} y_1 \\ \vdots \\ y_i \\ \vdots \\ y_m \end{pmatrix}, \quad \varepsilon = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_i \\ \vdots \\ \varepsilon_m \end{pmatrix}, \quad \gamma = \begin{pmatrix} \gamma_1 \\ \vdots \\ \gamma_i \\ \vdots \\ \gamma_m \end{pmatrix}$$

of all responses, errors, and random effects, respectively, as well as the design matrices

$$X = \begin{pmatrix} X_1 \\ \vdots \\ X_i \\ \vdots \\ X_m \end{pmatrix} \quad (7.19)$$

and

$$U = \text{blockdiag}(U_1, \dots, U_i, \dots, U_m) = \begin{pmatrix} U_1 & & & \mathbf{0} \\ & \ddots & & \\ & & U_i & \\ & & & \ddots \\ \mathbf{0} & & & & U_m \end{pmatrix}. \quad (7.20)$$

The LMM can then be written as

$$y = X\beta + U\gamma + \varepsilon$$

7.2 General Linear Mixed Model

Model

A general linear mixed model is given by

$$y = X\beta + U\gamma + \varepsilon$$

with

$$\begin{pmatrix} \gamma \\ \varepsilon \end{pmatrix} \sim N\left(\begin{pmatrix} \mathbf{0} \\ \mathbf{0} \end{pmatrix}, \begin{pmatrix} G & \mathbf{0} \\ \mathbf{0} & R \end{pmatrix}\right).$$

In this model, X and U are design matrices, β is a vector of fixed effects, and γ is a vector of random effects. The covariance matrices for γ and ε are assumed to be nonsingular, and therefore positive definite, and γ and ε are independent.

Software

Package `lme4` of R and `proc mixed` of SAS.

with $\varepsilon \sim N(\mathbf{0}, R)$, $\gamma \sim N(\mathbf{0}, G)$, and the block diagonal covariance matrices

$$\begin{aligned} R &= \text{blockdiag}(\sigma^2 \Sigma_{n_1}, \dots, \sigma^2 \Sigma_{n_i}, \dots, \sigma^2 \Sigma_{n_m}) \\ G &= \text{blockdiag}(Q, \dots, Q, \dots, Q). \end{aligned} \tag{7.21}$$

Note that ε and γ are still assumed to be independent. For i.i.d. errors, R simplifies to $R = \sigma^2 I$.

Box 7.2 summarizes LMMs in this compact notation. Such a representation is useful for two main reasons: First, likelihood inference (Sect. 7.3) and Bayesian inference (Sect. 7.4) can be similarly formulated. Secondly, and more importantly, statistical inference remains valid if we allow general design matrices X and U that are not of the special forms (7.19) and (7.20) and further allow more general covariance matrices G and R , which are not block-diagonal as in Eq. (7.21). We make primarily use of such general linear mixed models in Chaps. 8 and 9, representing non- and semiparametric regression models as mixed models. To give the reader a flavor of the capabilities of general LMMs, we briefly discuss multilevel models in the remarks that follow.

Remarks

1. *Multilevel models*: Mixed models for longitudinal or clustered data as described in Sect. 7.1 are suitable for modeling and analyzing data structures with one grouping level given by the individual or the cluster to which the observation

belongs. Multilevel models extend mixed models to data structures with more than one grouping level: Each observation may belong to several, possibly nested, groups. The classical example for this type of data structure results from school tests where pupils are grouped within classes and classes are grouped in schools. If we assume nested data, i.e., one pupil is member of exactly one class and each class belongs to exactly one school, then this leads to three-level models such as

$$y_{ijl} = \mathbf{x}'_{ijl}\boldsymbol{\beta} + \gamma_i + \gamma_{ij} + \varepsilon_{ijl},$$

where i indexes schools, j indexes classes, l indexes individual pupils within classes, and only random intercepts are considered. Naturally, random slopes can also be included at any level, and it is then of particular interest to compare effects of the same type of covariate on different levels. For example, the covariate “math score” can be constructed both on the class level as an average score and also on the individual level. Such three-level nested models allow for a variance decomposition on the different levels, similar to the presentation in mixed models (where it was limited to only two levels). Although the models have more than one grouping level, they can be written as a general LMM as defined in Box 7.2.

Multilevel models can also be extended to further hierarchical stages, as well as to non-nested models. Such models with more than two hierarchical levels will not be considered in this book, although they are useful in several areas of research, such as meta-analyses in clinical research or models for learning achievements. We recommend introductions provided in, for example, Gelman and Hill (2006), Skrondal and Rabe-Hesketh (2004), Scott, Simonoff, and Marx (2012), and Skrondal and Rabe-Hesketh (2008). The latter is an introduction with STATA but worth reading even if other software is used.

2. *Normality assumption:* As in LMMs for longitudinal and clustered data, the normality assumption is not necessary for all inferential results. However, estimation of unknown parameters in $\mathbf{G}$ and $\mathbf{R}$ is likelihood-based, thus we include the normality assumption in the definition.
3. *Conditional and marginal formulation:* We distinguish the conditional and the marginal model formulation. The conditional model for $\mathbf{y}$ given $\boldsymbol{\gamma}$ is

$$\mathbf{y} | \boldsymbol{\gamma} \sim \mathbf{N}(\mathbf{X}\boldsymbol{\beta} + \mathbf{U}\boldsymbol{\gamma}, \mathbf{R}),$$

together with the random effects distribution

$$\boldsymbol{\gamma} \sim \mathbf{N}(\mathbf{0}, \mathbf{G}).$$

The marginal model can be obtained by reexpressing the model in the form

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}^*, \quad \boldsymbol{\varepsilon}^* = \mathbf{U}\boldsymbol{\gamma} + \boldsymbol{\varepsilon}.$$

This results in the general marginal model

$$\mathbf{y} \sim \mathbf{N}(\mathbf{X}\boldsymbol{\beta}, \mathbf{R} + \mathbf{U}\mathbf{G}\mathbf{U}'), \quad (7.22)$$

or $\mathbf{y} \sim N(\mathbf{X}\boldsymbol{\beta}, \mathbf{V})$ with covariance matrix

$$\mathbf{V} = \mathbf{R} + \mathbf{U}\mathbf{G}\mathbf{U}'.$$

Note that the marginal model can be derived from the conditional model, but not vice versa. If interest only lies in estimation of fixed effects $\boldsymbol{\beta}$, the marginal model may be used. However, the conditional formulation is needed to estimate random effects $\boldsymbol{\gamma}$.

7.3 Likelihood Inference in LMMs

This section describes in Sects. 7.3.1–7.3.3 likelihood-based estimation of fixed and random effects, as well as of the unknown parameters in $\mathbf{G}$ and $\mathbf{R}$ in the general LMM of Box 7.2. Section 7.3.4 discusses tests of hypotheses. Bayesian inference is presented in Sect. 7.4.

7.3.1 Known Variance–Covariance Parameters

We first assume that all parameters in $\mathbf{R}$ and $\mathbf{G}$, and therefore in $\mathbf{V} = \mathbf{R} + \mathbf{U}\mathbf{G}\mathbf{U}'$, are known.

The unknown parameters can be estimated by simultaneously maximizing the joint log-likelihood of $\mathbf{y}, \boldsymbol{\gamma}$ with respect to $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$. The logarithm of $p(\mathbf{y}, \boldsymbol{\gamma}) = p(\mathbf{y} | \boldsymbol{\gamma})p(\boldsymbol{\gamma})$ yields the log-likelihood (up to an additive constant)

$$-\frac{1}{2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{U}\boldsymbol{\gamma})'\mathbf{R}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{U}\boldsymbol{\gamma}) - \frac{1}{2}\boldsymbol{\gamma}'\mathbf{G}^{-1}\boldsymbol{\gamma}.$$

It follows that maximization of the above expression is equivalent to minimizing

$$\text{LS}_{\text{pen}}(\boldsymbol{\beta}, \boldsymbol{\gamma}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{U}\boldsymbol{\gamma})'\mathbf{R}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{U}\boldsymbol{\gamma}) + \boldsymbol{\gamma}'\mathbf{G}^{-1}\boldsymbol{\gamma}. \quad (7.23)$$

The first term corresponds to a (general) *least squares criterion*. Without the second term, $\boldsymbol{\gamma}$ would be estimated exactly as a fixed effect. The second term takes into account that $\boldsymbol{\gamma}$ is random and follows a distribution, i.e., $\boldsymbol{\gamma} \sim N(\mathbf{0}, \mathbf{G})$, with positive definite covariance matrix $\mathbf{G}$. Additionally, $\boldsymbol{\gamma}'\mathbf{G}^{-1}\boldsymbol{\gamma}$ penalizes deviations of $\boldsymbol{\gamma}$ from $\mathbf{0}$ and is therefore called a penalty term. The penalty is large if $\mathbf{G}$ is “small,” and it is small if $\mathbf{G}$ is “large.” In the limiting case of $\mathbf{G}^{-1} \rightarrow \mathbf{0}$, the penalty term becomes 0, and $\boldsymbol{\gamma}$ is estimated exactly as a fixed effect without any distributional assumptions. For a random intercept model, the penalty Eq. (7.23) reduces to the form (7.7) where the quadratic penalty term shrinks parameters towards zero. See in Sect. 7.1.1 on p. 354f. for a more detailed discussion.

Setting the first derivatives of $\text{LS}_{\text{pen}}(\boldsymbol{\beta}, \boldsymbol{y})$ to zero results in the *mixed model equations*

$$\begin{pmatrix} X'R^{-1}X & X'R^{-1}U \\ U'R^{-1}X & U'R^{-1}U + G^{-1} \end{pmatrix} \begin{pmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\boldsymbol{y}} \end{pmatrix} = \begin{pmatrix} X'R^{-1}\boldsymbol{y} \\ U'R^{-1}\boldsymbol{y} \end{pmatrix}. \quad (7.24)$$

Defining $\boldsymbol{C} = (\boldsymbol{X}, \boldsymbol{U})$ and the partitioned matrix

$$\boldsymbol{B} = \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{G}^{-1} \end{pmatrix},$$

the solution of Eq. (7.24) can be written as

$$\begin{pmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\boldsymbol{y}} \end{pmatrix} = (\boldsymbol{C}'\boldsymbol{R}^{-1}\boldsymbol{C} + \boldsymbol{B})^{-1}\boldsymbol{C}'\boldsymbol{R}^{-1}\boldsymbol{y}. \quad (7.25)$$

This form also shows the close relationship to ridge estimation; see Sect. 4.2.2 (p. 203). A full derivation of the penalized least squares estimator Eq. (7.25) can be found in Sect. 7.8.2 on p. 412.

Some (tedious) matrix manipulations show that the estimator Eq. (7.25) has the form

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{V}^{-1}\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{V}^{-1}\boldsymbol{y} \quad (7.26)$$

and

$$\hat{\boldsymbol{y}} = \boldsymbol{G}\boldsymbol{U}'\boldsymbol{V}^{-1}(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}). \quad (7.27)$$

This implies that $\hat{\boldsymbol{\beta}}$ is a weighted least squares estimator with the inverse of the marginal covariance matrix $\boldsymbol{V} = \boldsymbol{R} + \boldsymbol{U}\boldsymbol{G}\boldsymbol{U}'$ of $\boldsymbol{\beta}$ as the weight matrix.

Omitting the normality assumption, $(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{y}})$ can also be derived as the best linear unbiased predictor (BLUP); see, for example, McCulloch and Searle (2001).

7.3.2 Unknown Variance–Covariance Parameters

This section describes the maximum likelihood (ML) and restricted maximum likelihood (REML) approach for estimating unknown parameters $\boldsymbol{\vartheta}$ in $\boldsymbol{R}$, $\boldsymbol{G}$, or $\boldsymbol{V}$. These estimators are the most widely used, and they are implemented in software such as SAS, STATA, and R. To make dependence on $\boldsymbol{\vartheta}$ explicit, we write

$$\boldsymbol{V} = \boldsymbol{V}(\boldsymbol{\vartheta}) = \boldsymbol{U}\boldsymbol{G}(\boldsymbol{\vartheta})\boldsymbol{U}' + \boldsymbol{R}(\boldsymbol{\vartheta}).$$

Maximum Likelihood Estimation of $\boldsymbol{\vartheta}$

ML estimation of $\boldsymbol{\vartheta}$ is based on the likelihood of the marginal model

$$\boldsymbol{y} \sim \text{N}(\boldsymbol{X}\boldsymbol{\beta}, \boldsymbol{V}(\boldsymbol{\vartheta})).$$

The corresponding log-likelihood, up to additive constants, is

$$\log L(\boldsymbol{\beta}, \boldsymbol{\vartheta}) = l(\boldsymbol{\beta}, \boldsymbol{\vartheta}) = -\frac{1}{2} [\log |V(\boldsymbol{\vartheta})| + (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})' V(\boldsymbol{\vartheta})^{-1} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})]. \quad (7.28)$$

The maximization of $l(\boldsymbol{\beta}, \boldsymbol{\vartheta})$ with respect to $\boldsymbol{\beta}$ (while holding $\boldsymbol{\vartheta}$ fixed) gives

$$\hat{\boldsymbol{\beta}}(\boldsymbol{\vartheta}) = (\mathbf{X}' V(\boldsymbol{\vartheta})^{-1} \mathbf{X})^{-1} \mathbf{X}' V(\boldsymbol{\vartheta})^{-1} \mathbf{y}.$$

Inserting $\hat{\boldsymbol{\beta}}(\boldsymbol{\vartheta})$ in $l(\boldsymbol{\beta}, \boldsymbol{\vartheta})$ results in the *profile log-likelihood*

$$l_P(\boldsymbol{\vartheta}) = -\frac{1}{2} [\log |V(\boldsymbol{\vartheta})| + (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}(\boldsymbol{\vartheta}))' V(\boldsymbol{\vartheta})^{-1} (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}(\boldsymbol{\vartheta}))].$$

The maximization of $l_P(\boldsymbol{\vartheta})$ with respect to $\boldsymbol{\vartheta}$ provides the ML estimator $\hat{\boldsymbol{\vartheta}}_{\text{ML}}$. Refer to [Harville \(1977\)](#) for explicit formulae and algorithms.

Restricted Maximum Likelihood Estimation of $\boldsymbol{\vartheta}$

Rather than using $l_P(\boldsymbol{\vartheta})$, estimation of $\boldsymbol{\vartheta}$ is often based on the *marginal or restricted log-likelihood*

$$l_R(\boldsymbol{\vartheta}) = \log \left(\int L(\boldsymbol{\beta}, \boldsymbol{\vartheta}) d\boldsymbol{\beta} \right),$$

integrating out $\boldsymbol{\beta}$ from the likelihood. This marginal log-likelihood is motivated from an empirical Bayesian perspective, where $\boldsymbol{\beta}$ is assumed to be random with a flat prior $p(\boldsymbol{\beta}) \propto \text{const}$. It can be shown that the restricted log-likelihood is

$$l_R(\boldsymbol{\vartheta}) = l_P(\boldsymbol{\vartheta}) - \frac{1}{2} \log |\mathbf{X}' V(\boldsymbol{\vartheta})^{-1} \mathbf{X}|, \quad (7.29)$$

and maximization of $l_R(\boldsymbol{\vartheta})$ provides the restricted ML estimator $\hat{\boldsymbol{\vartheta}}_{\text{REML}}$; see [Harville \(1974\)](#) for details.

As discussed in Sect. 3.2.2, the REML estimator

$$\hat{\sigma}^2 = \frac{1}{n-p} \sum_{i=1}^n (y_i - \mathbf{x}_i' \hat{\boldsymbol{\beta}})^2$$

removes the bias of the ML estimator for the error variance σ^2 in linear models. Just as in this simpler context, reduction of the bias of $\hat{\boldsymbol{\vartheta}}_{\text{ML}}$ is the main reason for preferring $\hat{\boldsymbol{\vartheta}}_{\text{REML}}$ in LMMs as an estimator for $\boldsymbol{\vartheta}$. However, in our more general setting, the REML estimator is generally *not* unbiased. A rather simple example for a biased REML estimator is the random intercept model with unbalanced data where the number of observations n_i is *different* in each cluster. Moreover, there are no general results available to ensure that the MSE is also reduced compared to full maximum likelihood.

The estimators $\hat{\boldsymbol{\theta}}_{\text{REML}}$ and $\hat{\boldsymbol{\theta}}_{\text{ML}}$ are computed numerically through iterative algorithms, for example, by maximizing $l_R(\boldsymbol{\theta})$ or $l_P(\boldsymbol{\theta})$ using Newton–Raphson or Fisher scoring algorithms; see Appendix B.4.2.

Finally, plugging in $\hat{\boldsymbol{\theta}}$ after convergence provides the estimated covariance matrices

$$\hat{\mathbf{R}} = \mathbf{R}(\hat{\boldsymbol{\theta}}), \quad \hat{\mathbf{G}} = \mathbf{G}(\hat{\boldsymbol{\theta}}), \quad \hat{\mathbf{V}} = \mathbf{V}(\hat{\boldsymbol{\theta}}).$$

Estimation of Fixed and Random Effects

The final estimators $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\gamma}}$ of fixed effects $\boldsymbol{\beta}$ and random effects $\boldsymbol{\gamma}$ are obtained using the estimates Eqs. (7.26) and (7.27) after replacing the matrices $\mathbf{R}$, $\mathbf{G}$, and $\mathbf{V}$ with their estimates $\hat{\mathbf{R}}$, $\hat{\mathbf{G}}$, and $\hat{\mathbf{V}}$. However, after plugging in $\hat{\boldsymbol{\theta}}$, optimality properties are no longer exactly valid.

A summary of likelihood inference in LMMs is provided in Box 7.3.

Random Intercept Model

As a special case, we consider the random intercept model

$$y_{ij} = \mathbf{x}'_{ij}\boldsymbol{\beta} + \gamma_{0i} + \varepsilon_{ij}, \quad i = 1, \dots, m, \quad j = 1, \dots, n_i,$$

or, equivalently,

$$\mathbf{y}_i = \mathbf{X}_i\boldsymbol{\beta} + \mathbf{U}_i\gamma_{0i} + \boldsymbol{\varepsilon}_i, \quad i = 1, \dots, m,$$

where $\boldsymbol{\varepsilon}_i \sim \mathbf{N}(\mathbf{0}, \sigma^2 \mathbf{I})$, $\gamma_{0i} \stackrel{i.i.d.}{\sim} \mathbf{N}(0, \tau_0^2)$, and $\mathbf{U}_i = (1, \dots, 1)'$ is a n_i -dimensional vector of ones.

Now $\mathbf{G} = \tau_0^2 \mathbf{I}_m$, $\mathbf{R} = \sigma^2 \mathbf{I}$, $\mathbf{V} = \text{blockdiag}(\mathbf{V}_1, \dots, \mathbf{V}_m)$ with

$$\begin{aligned} \mathbf{V}_i &= \sigma^2 \mathbf{I}_{n_i} + \tau_0^2 \mathbf{J}_{n_i}, \\ \mathbf{V}_i^{-1} &= \frac{1}{\sigma^2} \left(\mathbf{I}_{n_i} - \frac{\tau_0^2}{\sigma^2 + n_i \tau_0^2} \mathbf{J}_{n_i} \right). \end{aligned}$$

The inverse can be verified by direct multiplication. More specifically, the elements on the main diagonal of $\mathbf{V}_i^{-1}$ are given by

$$\frac{1}{\sigma^2(\sigma^2 + n_i \tau_0^2)},$$

and the elements above and below the main diagonal are $-\tau_0^2$. Therefore

$$\begin{aligned} \hat{\boldsymbol{\gamma}} &= \hat{\mathbf{G}} \mathbf{U}' \hat{\mathbf{V}}^{-1} (\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}) \\ &= \hat{\tau}_0^2 \text{blockdiag}(\mathbf{U}_1, \dots, \mathbf{U}_m)' \text{blockdiag}(\hat{\mathbf{V}}_1^{-1}, \dots, \hat{\mathbf{V}}_m^{-1}) (\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}). \end{aligned}$$

7.3 Likelihood Inference in LMMs

Estimation of Variance and Covariance Parameters

We denote $\boldsymbol{\vartheta}$ as the vector of unknown parameters in $\mathbf{R} = \mathbf{R}(\boldsymbol{\vartheta})$, $\mathbf{G} = \mathbf{G}(\boldsymbol{\vartheta})$, and $\mathbf{V} = \mathbf{V}(\boldsymbol{\vartheta})$. The ML estimator for $\boldsymbol{\vartheta}$ is obtained by maximizing the (profile) log-likelihood

$$l_P(\boldsymbol{\vartheta}) = -\frac{1}{2} \left[\log |\mathbf{V}(\boldsymbol{\vartheta})| + (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}(\boldsymbol{\vartheta}))' \mathbf{V}(\boldsymbol{\vartheta})^{-1} (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}(\boldsymbol{\vartheta})) \right]$$

with

$$\hat{\boldsymbol{\beta}}(\boldsymbol{\vartheta}) = (\mathbf{X}' \mathbf{V}(\boldsymbol{\vartheta})^{-1} \mathbf{X})^{-1} \mathbf{X}' \mathbf{V}(\boldsymbol{\vartheta})^{-1} \mathbf{y}.$$

The REML (restricted ML) estimator is obtained by maximizing the restricted log-likelihood

$$l_R(\boldsymbol{\vartheta}) = l_P(\boldsymbol{\vartheta}) - \frac{1}{2} \log |\mathbf{X}' \mathbf{V}(\boldsymbol{\vartheta})^{-1} \mathbf{X}|.$$

Estimation of Fixed and Random Effects

Estimators $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\gamma}}$ for fixed and random effects are

$$\begin{aligned} \hat{\boldsymbol{\beta}} &= (\mathbf{X}' \hat{\mathbf{V}}^{-1} \mathbf{X})^{-1} \mathbf{X}' \hat{\mathbf{V}}^{-1} \mathbf{y} \\ \hat{\boldsymbol{\gamma}} &= \hat{\mathbf{G}} \mathbf{U}' \hat{\mathbf{V}}^{-1} (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}), \end{aligned}$$

or, equivalently,

$$\begin{pmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\boldsymbol{\gamma}} \end{pmatrix} = (\mathbf{C}' \hat{\mathbf{R}}^{-1} \mathbf{C} + \hat{\mathbf{B}})^{-1} \mathbf{C}' \hat{\mathbf{R}}^{-1} \mathbf{y},$$

with $\mathbf{C} = (\mathbf{X}, \mathbf{U})$ and $\hat{\mathbf{B}} = \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \hat{\mathbf{G}}^{-1} \end{pmatrix}$. These estimators are also referred to as empirical best linear predictors (EBLP or EBLUP).

Estimated Covariance Matrix

Bayesian Covariance Matrix

$$\widehat{\text{Cov}} \begin{pmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\boldsymbol{\gamma}} \end{pmatrix} = \mathbf{A} = (\mathbf{C}' \hat{\mathbf{R}}^{-1} \mathbf{C} + \hat{\mathbf{B}})^{-1}$$

Frequentist Covariance Matrix

$$\widehat{\text{Cov}} \left(\begin{pmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\boldsymbol{\gamma}} \end{pmatrix} \mid \mathbf{y} \right) = \mathbf{A} \mathbf{C}' \hat{\mathbf{R}}^{-1} \mathbf{C} \mathbf{A}$$

Table 7.2 Hormone therapy with rats: comparison of estimates based on REML and ML

	Parameter	REML		ML	
		Estimate	s.e.	Estimate	s.e.
<i>intercept</i>	β_0	68.606	0.327	68.607	0.326
<i>low-dose</i>	β_1	7.505	0.227	7.508	0.224
<i>high-dose</i>	β_2	6.875	0.230	6.869	0.226
<i>control</i>	β_3	7.318	0.284	7.315	0.279
$\text{Var}(\gamma_{0i})$	τ_0^2	3.430		3.417	
$\text{Var}(\gamma_{1i})$	τ_1^2	0.001		<0.001	
$\text{Var}(\varepsilon_{ij})$	σ^2	1.444		1.428	

Hence, the estimated random intercepts are

$$\hat{\gamma}_{0i} = \hat{\tau}_0^2 (1, \dots, 1) \hat{V}_i^{-1} (\mathbf{y}_i - \mathbf{X}_i \hat{\boldsymbol{\beta}}),$$

which can be simplified to

$$\hat{\gamma}_{0i} = \frac{n_i \hat{\tau}_0^2}{\hat{\sigma}^2 + n_i \hat{\tau}_0^2} \left\{ \frac{1}{n_i} \sum_{j=1}^{n_i} (y_{ij} - \mathbf{x}'_{ij} \hat{\boldsymbol{\beta}}) \right\} = \frac{n_i \hat{\tau}_0^2}{\hat{\sigma}^2 + n_i \hat{\tau}_0^2} e_i$$

with the average residual $e_i = \frac{1}{n_i} \sum_{j=1}^{n_i} (y_{ij} - \mathbf{x}'_{ij} \hat{\boldsymbol{\beta}})$. An interpretation of this result is already given in Sect. 7.1.1 on p. 354ff.

Example 7.3 Hormone Therapy with Rats—Estimated Effects

The results presented in Example 7.1 (p. 359) are obtained using REML to estimate the variance and covariance parameters of random effects. REML is also the default in most statistical software packages when estimating mixed models. Table 7.2 contrasts the REML estimates with maximum likelihood for the random slope model (7.12). Generally the results are very close. The estimates for the variance components with ML are slightly below those based on REML. This could have been expected as ML is more biased downwards towards zero compared to REML, i.e., it tends to underestimate the variances.

The estimated random effects $\hat{\gamma}_{0i}$ and $\hat{\gamma}_{1i}$ of model (7.12) are best presented graphically. A particularly useful visual representation of the random effects is found in *caterpillar plots*. These plots show the estimated random effects in increasing order together with 95 % confidence intervals; see the next section for the computation of the latter. Figure 7.5 shows such caterpillar plots for model (7.12). Clearly, the estimated random intercept parameters $\hat{\gamma}_{0i}$ show considerable variability indicating large rat-specific deviations from the average height of skull. On the other hand, the estimated random slope parameters $\hat{\gamma}_{1i}$ are all estimated close to zero (note the different scaling of the figures). This is not really a surprise as already the estimated random slope variance is essentially zero; refer to Table 7.2.

Another option to visualize the random effects distribution is to produce (nonparametric) kernel densities computed from the estimates $\hat{\gamma}_{0i}$ and $\hat{\gamma}_{1i}$, $i = 1, \dots, 50$, of the random effects. Such plots are also helpful to investigate the normality assumption of the random

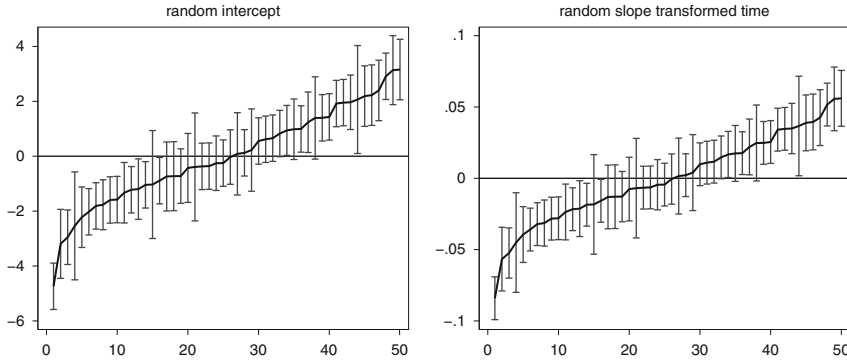


Fig. 7.5 Hormone therapy with rats: caterpillar plots of the estimated random effects $\hat{\gamma}_{0i}$ and $\hat{\gamma}_{1i}$ in the random slope model (7.12)

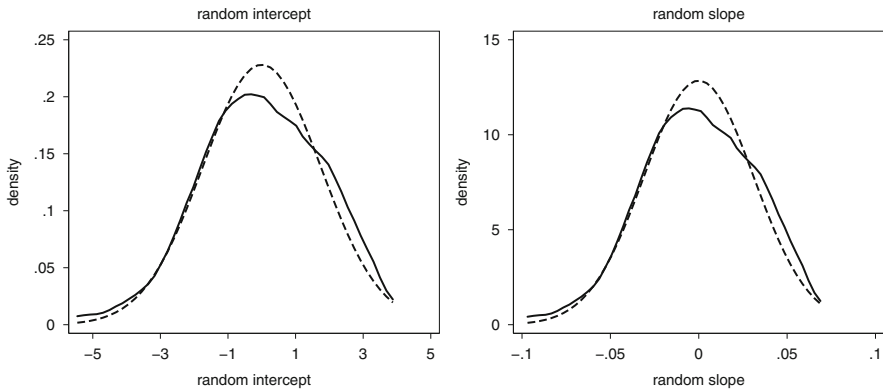


Fig. 7.6 Hormone therapy with rats: kernel density estimators (solid line) and normal approximation (dashed line) for the random effects

effects (at least to a certain extent). Figure 7.6 shows kernel densities for $\hat{\gamma}_{0i}$ and $\hat{\gamma}_{1i}$. Deviations from the assumed normal distribution are comparably small. Once again, the random slopes are close to zero.

The estimated random effects can also be used to obtain estimates for the individual (rat-specific) trends. These trends are computed by inserting the estimates into the trend lines. For instance, for the low-dose group, we obtain the estimated trends $68.606 + \hat{\gamma}_{0i} + 7.505t_{ij} + \hat{\gamma}_{1i}t_{ij}$. Plots of the individual trends are given in Fig. 7.7 which again reveals considerable rat-specific heterogeneity at least regarding the intercept. The slopes of the individual trends are almost in parallel. This is another indicator that the simple random intercept model (7.13) is sufficient. In fact, if we would superimpose the trends obtained from Eq. (7.13), they would be visually indistinguishable.

△

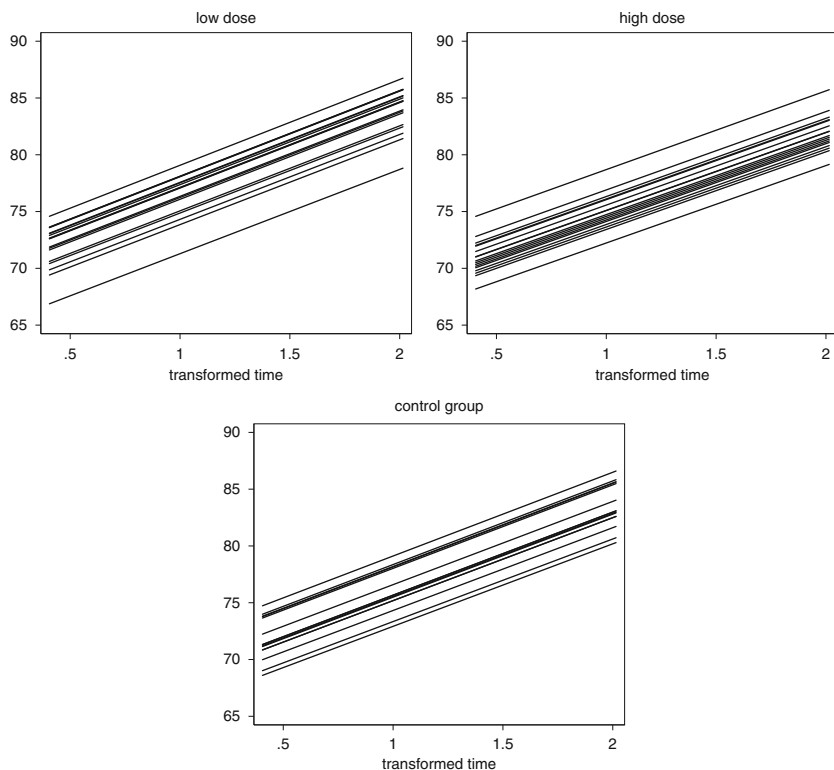


Fig. 7.7 Hormone therapy with rats: individual estimated trends in the random slope model (7.12)

7.3.3 Variability of Fixed and Random Effects Estimators

Bayesian Covariance Matrix

For known variance and covariance parameters, we can derive the covariance matrix

$$\text{Cov} \begin{pmatrix} \hat{\beta} \\ \hat{\gamma} \end{pmatrix} = (C' R^{-1} C + B)^{-1} \quad (7.30)$$

from a Bayesian perspective; see Sect. 7.4.1 (p. 384) for a derivation. Plugging in $\hat{R}$ and $\hat{B}$, we obtain the estimated covariance matrix

$$\widehat{\text{Cov}} \begin{pmatrix} \hat{\beta} \\ \hat{\gamma} \end{pmatrix} = A = (C' \hat{R}^{-1} C + \hat{B})^{-1}. \quad (7.31)$$

Note that the additional variability introduced by plugging in estimates $\hat{\mathbf{R}}$ and $\hat{\mathbf{B}}$ for $\mathbf{R}$ and $\mathbf{B}$ is not taken into account in Eq. (7.31), implying a tendency to underestimate the true covariance matrix.

To construct confidence intervals or confidence bands for $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\gamma}}$, it is assumed that $(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\gamma}})$ has an approximate normal distribution for large sample size. For special LMMs, asymptotic results can be derived under appropriate regularity assumptions (e.g., Ruppert et al., 2003). However, generally, rigorous proofs remain open problems. Based on the normality approximation, quantiles and confidence intervals for the components of $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\gamma}}$ can be computed.

Frequentist Covariance Matrix

An alternative approximate covariance matrix can be derived by conditioning on $\boldsymbol{\gamma}$. Then $\mathbf{y} | \boldsymbol{\gamma} \sim N(\mathbf{X}\boldsymbol{\beta} + \mathbf{U}\boldsymbol{\gamma}, \mathbf{R})$ and $\text{Cov}(\mathbf{y} | \boldsymbol{\gamma}) = \mathbf{R}$. Applying the rules for linear transformations (Appendix B.2, Theorem B.2.5) to Eq. (7.25) yields

$$\text{Cov}\left(\begin{pmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\boldsymbol{\gamma}} \end{pmatrix} | \boldsymbol{\gamma}\right) = (\mathbf{C}'\mathbf{R}^{-1}\mathbf{C} + \mathbf{B})^{-1}\mathbf{C}'\mathbf{R}^{-1}\mathbf{C}(\mathbf{C}'\mathbf{R}^{-1}\mathbf{C} + \mathbf{B})^{-1}. \quad (7.32)$$

This covariance matrix has the form of a “sandwich” matrix, composed of the terms $(\mathbf{C}'\mathbf{R}^{-1}\mathbf{C} + \mathbf{B})^{-1}$ and $\mathbf{C}'\mathbf{R}^{-1}\mathbf{C}$ (see also Sects. 4.1.3, p. 190, 8.1.2, p. 439, and Sect. 8.1.9, p. 486). Inserting the estimates $\hat{\mathbf{R}}$ and $\hat{\mathbf{B}}$ provides the estimated covariance matrix

$$\widehat{\text{Cov}}\left(\begin{pmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\boldsymbol{\gamma}} \end{pmatrix} | \boldsymbol{\gamma}\right) = \mathbf{A}\mathbf{C}'\hat{\mathbf{R}}^{-1}\mathbf{C}\mathbf{A}, \quad (7.33)$$

where $\mathbf{A}$ is the Bayesian covariance matrix Eq. (7.31). In comparison to $\mathbf{A}$, Eq. (7.33) tends to further underestimate the true covariance matrix because the additional factor $\mathbf{C}'\hat{\mathbf{R}}^{-1}\mathbf{C}\mathbf{A}$ is “smaller” than the identity matrix $\mathbf{I}$. Therefore, we prefer the Bayesian covariance matrix Eq. (7.31). Motivated by its derivation, Eq. (7.33) is called the frequentist version of the covariance matrix.

Approximate Covariance Matrices for Covariance Parameters

Approximate covariance matrices $\widehat{\text{Cov}}(\hat{\boldsymbol{\vartheta}})$ for the estimators $\hat{\boldsymbol{\vartheta}}_{\text{ML}}$ and $\hat{\boldsymbol{\vartheta}}_{\text{REML}}$ can be obtained from the Fisher information involved in Newton–Raphson or Fisher scoring iterations. However, since the estimators are skewed, particularly for variances, these approximations are relatively imprecise and are less appropriate for constructing confidence intervals or tests.

7.3.4 Testing Hypotheses

Testing Fixed Effects Parameters

Hypotheses on fixed effects are often of primary interest. The simplest case is testing

$$H_0 : \beta_j = d_j \quad \text{versus} \quad H_1 : \beta_j \neq d_j$$

for a component β_j of $\boldsymbol{\beta}$.

With $d_j = 0$, this is a test on the significance of the j th covariate, just as in the linear regression model. Assuming an approximate normal distribution for $\hat{\boldsymbol{\beta}}$, a decision can be based on an (approximate) confidence interval for β_j . H_0 is rejected at a significance level α if d_j is outside the corresponding $1 - \alpha$ confidence interval. Equivalently, the test statistic

$$t_j = \frac{\hat{\beta}_j - d_j}{\hat{\sigma}_j}$$

can be used, where $\hat{\sigma}_j$ is the square root of the corresponding diagonal element of the approximate covariance matrix Eq. (7.31) of $\hat{\boldsymbol{\beta}}$, $\hat{\boldsymbol{\gamma}}$. Under normality assumptions on $\boldsymbol{\gamma}$ and $\boldsymbol{\varepsilon}$, the test statistic t_j even has an exact t-distribution (under H_0) in special cases (e.g., Ruppert et al., 2003). In general, it is assumed that t_j is approximately standard normal under the null hypothesis, and H_0 is rejected if $|t_j| > z_{1-\alpha/2}$.

To test linear hypotheses

$$H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d} \quad \text{versus} \quad H_1 : \mathbf{C}\boldsymbol{\beta} \neq \mathbf{d}$$

one may consider the Wald statistic (see Appendix B.4.4)

$$W = (\mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d})'(\mathbf{C}\widehat{\text{Cov}}(\hat{\boldsymbol{\beta}})\mathbf{C}')^{-1}(\mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d}),$$

where $\widehat{\text{Cov}}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}'\hat{\mathbf{V}}^{-1}\mathbf{X})^{-1}$ is the estimated covariance matrix of $\hat{\boldsymbol{\beta}}$. However, $(\mathbf{X}'\hat{\mathbf{V}}^{-1}\mathbf{X})^{-1}$ might be a poor estimate of $\text{Cov}(\hat{\boldsymbol{\beta}})$ for small samples, and the asymptotic χ^2 -distribution for W is lacking a general rigorous foundation. Therefore Wald tests, or F-tests with $F = W/\text{rk}(\mathbf{C})$ as implemented in SAS, STATA, and R, must be used with caution.

Alternatively, tests could be carried out with the likelihood ratio statistic

$$\text{LRT} = 2\{l(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\vartheta}}) - l(\tilde{\boldsymbol{\beta}}, \tilde{\boldsymbol{\vartheta}})\}$$

or with the restricted likelihood ratio statistic

$$\text{RLRT} = 2\{l_R(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\vartheta}}) - l_R(\tilde{\boldsymbol{\beta}}, \tilde{\boldsymbol{\vartheta}})\},$$

where $l(\boldsymbol{\beta}, \boldsymbol{\vartheta})$ is the log-likelihood Eq. (7.28) of the marginal model and $l_R(\boldsymbol{\beta}, \boldsymbol{\vartheta}) = l(\boldsymbol{\beta}, \boldsymbol{\vartheta}) - 0.5 \log |\mathbf{X}'\mathbf{V}(\boldsymbol{\vartheta})^{-1}\mathbf{X}|$ is the restricted log-likelihood. Thereby $(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\vartheta}})$ are

the unrestricted ML estimates and $(\tilde{\beta}, \tilde{\vartheta})$ are the ML estimates under H_0 . However, cautionary remarks have to be made for these tests, as outlined in [Crainiceanu and Ruppert \(2004\)](#).

An important example for a general linear hypothesis in the context of LMMs is the test for equality of β_1 and β_2 in the extended random intercept model

$$y_{ij} = \beta_0 + \beta_1(x_{ij} - \bar{x}_i) + \beta_2\bar{x}_i + \gamma_{0i} + \varepsilon_{ij},$$

discussed in Sect. 7.1.1. Equality of β_1 and β_2 means that the between- and within-cluster effects of x are identical. If $H_0 : \beta_1 = \beta_2$ is rejected, there are significantly different within- and between-cluster effects. The test can also be seen as a test for correlation between random effects and observed covariates or endogeneity; see the discussion in Sect. 7.1.5 on stochastic covariates. A rejection of H_0 then provides evidence of correlation between random effects and covariates or endogenous x . In the econometrics literature testing endogeneity is usually performed using the (among econometricians) well-known Hausman specification test; see [Hausmann \(1978\)](#). In the context of the random intercept model the Hausman test is equivalent to the Wald test described above.

Testing Random Effects or Variance Parameters

Testing hypotheses about random effects or random effects variances is more difficult and generally applicable tests are comparably scarce. Consider, for example, the random intercept model

$$y_{ij} = \beta_0 + \beta_1 x_{ij} + \gamma_{0i} + \varepsilon_{ij}, \quad i = 1, \dots, m, \quad j = 1, \dots, n_i,$$

with $\varepsilon_{ij} \sim N(0, \sigma^2)$ and $\gamma_{0i} \sim N(0, \tau_0^2)$. The null hypothesis $H_0 : \tau_0^2 = 0$ is then equivalent to $H_0 : \boldsymbol{\gamma} = (\gamma_{01}, \dots, \gamma_{0m})' = \mathbf{0}$ and the simple linear regression model

$$y_{ij} = \beta_0 + \beta_1 x_{ij} + \varepsilon_{ij}$$

without the random intercept γ_{0i} .

Extending the random intercept model to a random slope model

$$y_{ij} = \beta_0 + \beta_1 x_{ij} + \gamma_{0i} + \gamma_{1i} x_{ij} + \varepsilon_{ij}$$

with $\gamma_{1i} \sim N(0, \tau_1^2)$, the null hypothesis $H_0 : \tau_1^2 = 0$ or $H_0 : \boldsymbol{\gamma}_1 = (\gamma_{11}, \dots, \gamma_{1m})' = \mathbf{0}$ is then equivalent to the simpler random intercept model.

In SAS, an F-test is implemented for testing linear hypotheses about random effects vectors, such as $H_0 : \boldsymbol{\gamma}_0 = \mathbf{0}$ or $H_0 : \boldsymbol{\gamma}_1 = \mathbf{0}$. However, we do not encourage its use because it can be quite conservative and its power is weak. Of course, if a null hypothesis such as $\boldsymbol{\gamma}_0 = \mathbf{0}$ or $\boldsymbol{\gamma}_1 = \mathbf{0}$ is rejected, we can safely include the random effect component. On the other hand, if H_0 is not rejected, there is a considerable risk to falsely omit the random effect component.

As an alternative, (restricted) likelihood ratio tests for testing hypotheses on random effects variances, such as

$$H_0 : \tau_0^2 = 0 \quad \text{versus} \quad H_1 : \tau_0^2 > 0 \quad (7.34)$$

or

$$H_0 : \tau_1^2 = 0 \quad \text{versus} \quad H_1 : \tau_1^2 > 0 \quad (7.35)$$

should be considered, based on the LRT or, even better, the RLRT statistics. For the random intercept model, the parameters are $\beta = (\beta_0, \beta_1)'$ and $\boldsymbol{\vartheta} = (\sigma^2, \tau_0^2)$, with $\tau_0^2 = 0$ under H_0 , while β_0, β_1 , and σ^2 are “nuisance parameters.” For the random slope model, $\boldsymbol{\vartheta} = (\sigma^2, \tau_0^2, \tau_1^2)$, with $\tau_1^2 = 0$ under H_0 , and $\beta_0, \beta_1, \sigma^2$ and τ_0^2 are “nuisance parameters.”

However, standard asymptotic theory is violated for tests of this form, because $\tau_0^2 = 0$ or $\tau_1^2 = 0$ is on the boundary of the parameter space. [Stram and Lee \(1994\)](#) show that the asymptotic distribution of LRT and RLRT is a $0.5\chi_0^2 : 0.5\chi_1^2$ mixture distribution, where χ_0^2 denotes a point mass in zero. The latter has to be included because, with a chance of (asymptotically) 50 %, the random effects variance is estimated to be zero under the null hypothesis. Testing the variance of a random effects component based on this asymptotic mixture distribution is implemented, for example, in STATA. However, it has to be applied with caution because of the assumptions made by [Stram and Lee \(1994\)](#). They assume that the response vector $\mathbf{y}$ can be divided into subvectors $\mathbf{y}_i$, $i = 1, \dots, m$, which are independent and identically distributed (in the marginal model) under both the null hypothesis and the alternative. Such a decomposition is possible for random intercept models with a balanced design $n_i = n$, i.e., with the same number of observations within each cluster or for each individual. It is not possible, however, for unbalanced designs or for testing Eq. (7.35) in random slope models in the presence of a random intercept. Moreover, it is required that the number m of clusters tends to infinity. This is an appropriate assumption for longitudinal data with a large number m of individuals, but not for clustered data with a small number of clusters, even if the numbers of observations within clusters are large. [Giampaoli and Singer \(2009\)](#) relax the quite restrictive assumptions of [Stram and Lee \(1994\)](#) and prove that the (R)LRT statistic for testing Eq. (7.34) still has an asymptotic $0.5\chi_0^2 : 0.5\chi_1^2$ distribution, without requiring balanced data or large m . They also consider the hypothesis Eq. (7.35) for testing one random slope in the presence of a random intercept. In this case, however, the asymptotic distribution of (R)LRT is a $0.5\chi_1^2 : 0.5\chi_2^2$ mixture. It seems tempting to extend this result to models with more random slope components (as was done in [Stram and Lee \(1994\)](#) under the stronger assumptions discussed above), but no corresponding results are available.

[Crainiceanu and Ruppert \(2004a\)](#) and [Crainiceanu, Ruppert, Claeskens, and Wand \(2005\)](#) have investigated the finite sample and asymptotic distributions of the (R)LRT statistic under various scenarios and found that finite sample deviations from the $0.5\chi_0^2 : 0.5\chi_1^2$ mixture can be considerable.

However, Crainiceanu and Ruppert (2004a) also present representations of the finite sample distribution of (R)LRT tests that allow to construct exact tests, based on simulations, for zero variances in LMMs with one variance component. Scheipl, Greven, and Küchenhoff (2008) compare (R)LRT tests and F-type tests in a massive simulation study, concluding that RLRT tests are superior to F-tests. They include a fast and reliable approximation of the exact RLRT test and an extension to LMMs with several variance components, implemented in the R package `RLRsim`. It is recommended to apply it at least in addition to the tests based on asymptotic χ^2 -mixtures.

We summarize as follows: For testing linear hypotheses on fixed effects, the same toolkit of test statistics (t-, F-, Wald, and (restricted) likelihood ratio test statistics) is available as for linear and generalized linear models. Conclusions should be drawn with more caution, however, in particular for more complex models involving many parameters and random effects. The main reason is a lack of rigorous asymptotic justification, caused by correlation of observations induced by random effects.

In comparison, reliable tests to select or deselect random effects components are more scarce. This is not surprising because these tests are faced with non-standard problems, not covered by classical testing theory. For random intercept models, the $0.5\chi_0^2 : 0.5\chi_1^2$ mixture test is a sound asymptotic approximation. In addition, the test based on the exact finite sample distribution in the R package `RLRsim` should be applied. For models with one (scalar) additional random slope, the $0.5\chi_1^2 : 0.5\chi_2^2$ mixture test is still justified asymptotically. In all other cases, with more random slopes, only the test offered in `RLRsim` seems to be a reliable approximation. A good strategy is always to compare results with and without certain random slopes.

We finally note that there are some recent Bayesian approaches for random effects selection; see, e.g., Chen and Dunson (2003) and Frühwirth-Schnatter and Tüchler (2008).

7.4 Bayesian Linear Mixed Models

From a Bayesian perspective, both the population effects β and the random effects γ in a LMM

$$y = X\beta + U\gamma + \varepsilon$$

are considered to be random variables. The priors for β and γ are different, however. A *noninformative* prior

$$p(\beta) \propto \text{const}$$

corresponds to the frequentist approach, where β is assumed to be fixed but (completely) unknown. A more informative prior is, for example, a Gaussian prior

$$\beta \sim N(m, M),$$

where m and M are assumed to be known. The noninformative, flat prior $p(\beta) \propto \text{const}$ is obtained for the limiting case $M^{-1} \rightarrow \mathbf{0}$, i.e., when the precision matrix

tends to zero. The choice of a Gaussian prior is also motivated by the fact that, as in linear regression models, posterior inference is relatively straightforward. In a Bayesian approach, the random effects distribution

$$\boldsymbol{\gamma} \sim N(\mathbf{0}, \mathbf{G})$$

in the general LMM of Sect. 7.2 is now seen as a prior for the random effects. However, the covariance matrices $\mathbf{G}$ and $\mathbf{R}$ will usually contain a vector $\boldsymbol{\vartheta}$ of unknown hyperparameters. To make explicit dependence on these hyperparameters, we write

$$\boldsymbol{\gamma} \sim N(\mathbf{0}, \mathbf{G}(\boldsymbol{\vartheta})), \quad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \mathbf{R}(\boldsymbol{\vartheta})).$$

In addition, $\boldsymbol{\beta}$, $\boldsymbol{\gamma}$, and $\boldsymbol{\varepsilon}$ are assumed to be independent a priori.

Two different concepts exist for Bayesian inference, including the parameter $\boldsymbol{\vartheta}$: In a *fully Bayesian* approach, $\boldsymbol{\vartheta}$ is also considered as a random variable with a hyperprior $p(\boldsymbol{\vartheta})$. In contrast, $\boldsymbol{\vartheta}$ is considered as unknown, but fixed, in an *empirical Bayes* approach and is estimated, for example, through maximization of the (marginal) likelihood. The estimate $\hat{\boldsymbol{\vartheta}}$ is then inserted into $\mathbf{G}(\boldsymbol{\vartheta})$ and $\mathbf{R}(\boldsymbol{\vartheta})$, and Bayesian inference for $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ proceeds as with “known” covariance matrices $\hat{\mathbf{G}} = \mathbf{G}(\hat{\boldsymbol{\vartheta}})$ and $\hat{\mathbf{R}} = \mathbf{R}(\hat{\boldsymbol{\vartheta}})$.

7.4.1 Estimation for Known Covariance Structure

We first consider the *posterior distribution* of $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ for known (or given) covariance matrices $\mathbf{G}$ and $\mathbf{R}$, required in both Bayesian approaches. Applying Bayes’ theorem, the *joint posterior* is determined through

$$\begin{aligned} p(\boldsymbol{\beta}, \boldsymbol{\gamma} | \mathbf{y}) &\propto p(\mathbf{y} | \boldsymbol{\beta}, \boldsymbol{\gamma}) p(\boldsymbol{\beta}) p(\boldsymbol{\gamma}) \\ &\propto \exp\left(-\frac{1}{2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{U}\boldsymbol{\gamma})' \mathbf{R}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{U}\boldsymbol{\gamma})\right) \\ &\quad \cdot \exp\left(-\frac{1}{2}(\boldsymbol{\beta} - \mathbf{m})' \mathbf{M}^{-1}(\boldsymbol{\beta} - \mathbf{m})\right) \cdot \exp\left(-\frac{1}{2}\boldsymbol{\gamma}' \mathbf{G}^{-1}\boldsymbol{\gamma}\right). \end{aligned}$$

Defining

$$\mathbf{C} = (\mathbf{X}, \mathbf{U}), \quad \mathbf{B} = \begin{pmatrix} \mathbf{M}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{G}^{-1} \end{pmatrix}, \quad \tilde{\mathbf{m}} = \begin{pmatrix} \mathbf{M}^{-1}\mathbf{m} \\ \mathbf{0} \end{pmatrix},$$

algebraic manipulations (similar as for Bayesian linear models in Sect. 4.4) show that the posterior is a multivariate Gaussian distribution with mean vector

$$\begin{pmatrix} \hat{\beta} \\ \hat{\gamma} \end{pmatrix} = E \left(\begin{pmatrix} \beta \\ \gamma \end{pmatrix} \mid y \right) = (C' R^{-1} C + B)^{-1} (\tilde{m} + C' R^{-1} y) \quad (7.36)$$

and covariance matrix

$$\text{Cov} \left(\begin{pmatrix} \beta \\ \gamma \end{pmatrix} \mid y \right) = (C' R^{-1} C + B)^{-1}.$$

For the case of a noninformative prior $p(\beta) \propto \text{const}$, we can simply set $M^{-1} = \mathbf{0}$, resulting in

$$B = \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & G^{-1} \end{pmatrix}, \quad \tilde{m} = \begin{pmatrix} \mathbf{0} \\ \mathbf{0} \end{pmatrix}.$$

The posterior mean then coincides with the BLUP estimator (7.25), while the posterior covariance matrix is the Bayesian covariance matrix (7.30).

7.4.2 Estimation for Unknown Covariance Structure

Empirical Bayes Estimation

In an empirical Bayes approach the covariance parameters ϑ are considered as unknown, but fixed, and are estimated by maximizing the (marginal) likelihood

$$p(y \mid \vartheta) \propto \int p(y \mid \beta, \gamma, \vartheta) p(\gamma) p(\beta) d\beta d\gamma \quad (7.37)$$

with respect to ϑ . For a noninformative prior $p(\beta) \propto \text{const}$, it can be shown that $\log p(y \mid \vartheta) = l_R(\vartheta)$, i.e., the logarithm of the marginal likelihood coincides with the restricted log-likelihood (7.29). Thus, for a noninformative prior for β , the empirical Bayes estimator of ϑ is identical with the REML estimator $\hat{\vartheta}_{\text{REML}}$.

The empirical Bayes estimator for β and γ results if the covariance matrices G and R in Eq. (7.36) are replaced by their estimates $\hat{G}$ and $\hat{R}$. For a noninformative prior for β , the empirical Bayes estimator is identical with the empirical best linear predictor ($\hat{\beta}, \hat{\gamma}$) of Box 7.3.

Fully Bayesian Estimation

In a fully Bayesian approach, a prior $p(\vartheta)$ must be specified for the unknown parameters ϑ in $G = G(\vartheta)$ and $R = R(\vartheta)$. Inference is then based on the posterior

$$p(\beta, \gamma, \vartheta \mid y) \propto p(y \mid \beta, \gamma, \vartheta) p(\beta) p(\gamma \mid \vartheta) p(\vartheta),$$

with β, γ and ϑ assumed to be independent. To ensure that $p(\beta, \gamma, \vartheta \mid y)$ is a proper posterior density,

$$p(y) = \int p(y | \beta, \gamma, \vartheta) p(\beta) p(\gamma | \vartheta) p(\vartheta) d\beta d\gamma d\vartheta < \infty \quad (7.38)$$

must hold for the marginal density in the denominator of Bayes' theorem. If $p(\vartheta)$ is a proper, informative prior, i.e., $\int p(\vartheta) d\vartheta = 1$, then $p(\beta, \gamma, \vartheta | y)$ is also proper. For an improper prior with $\int p(\vartheta) d\vartheta = \infty$, the posterior is not generally guaranteed to be proper.

In general, the integral in Eq. (7.38) cannot be solved analytically, so that the posterior $p(\beta, \gamma, \vartheta | y)$ is not available in closed form. Therefore, fully Bayesian inference typically relies on MCMC simulation, with the entire vector $(\beta, \gamma, \vartheta)$ of parameters partitioned into subvectors. As long as no specific covariance structures are assumed, these subvectors (or blocks) are β , γ , and ϑ . The full conditional distributions of each subvector, given the data y and the rest of parameters, then have to be derived so that random numbers can be drawn sequentially; see Appendix B.5.

Similarly to the classical linear model in Sect. 4.4, it can be shown that the full conditionals for β and γ are multivariate Gaussian. For β , we obtain $\beta | \cdot \sim N(\mu_\beta, \Sigma_\beta)$ with

$$\mu_\beta = (X' R(\vartheta)^{-1} X + M^{-1})^{-1} (M^{-1} m + X' R(\vartheta)^{-1} (y - U\gamma)) \quad (7.39)$$

$$\Sigma_\beta = (X' R(\vartheta)^{-1} X + M^{-1})^{-1}. \quad (7.40)$$

The full conditional for γ is given by $\gamma | \cdot \sim N(\mu_\gamma, \Sigma_\gamma)$ with

$$\mu_\gamma = (U' R(\vartheta)^{-1} U + G^{-1})^{-1} (U' R(\vartheta)^{-1} (y - X\beta))$$

$$\Sigma_\gamma = (U' R(\vartheta)^{-1} U + G^{-1})^{-1}.$$

With a noninformative prior $p(\beta) \propto \text{const}$, we have $M^{-1} = \mathbf{0}$ and the conditional mean

$$\mu_\beta = (X' R(\vartheta)^{-1} X)^{-1} X' R(\vartheta)^{-1} (y - U\gamma)$$

is the weighted least squares estimator applied to $y - U\gamma$, i.e., the data y adjusted for the random effects $U\gamma$. In analogy, μ_γ is a Bayes estimator with prior $\gamma \sim N(\mathbf{0}, G(\vartheta))$, applied to $y - X\beta$.

The full conditional $p(\vartheta | \cdot)$ can be simplified for special LMMs with appropriately specified prior $p(\vartheta)$. As an example, we consider the Bayesian LMM for longitudinal or clustered data

$$y_{ij} = \mathbf{x}'_{ij} \beta + \mathbf{u}'_{ij} \gamma_i + \varepsilon_{ij}, \quad i = 1, \dots, m, \quad j = 1, \dots, n_i,$$

with i.i.d. errors ε_{ij} . It follows that $\text{Cov}(\varepsilon_i) = \sigma^2 \mathbf{I}_{n_i}$, $\mathbf{R} = \sigma^2 \mathbf{I}$, $\text{Cov}(\gamma_i) = \mathbf{Q}$, $\mathbf{G} = \text{blockdiag}(\mathbf{Q}, \dots, \mathbf{Q})$, and

$$X = \begin{pmatrix} X_1 \\ \vdots \\ X_i \\ \vdots \\ X_m \end{pmatrix} \quad U = \begin{pmatrix} U_1 & & \mathbf{0} \\ & \ddots & \\ & & U_i & \\ & & & \ddots \\ \mathbf{0} & & & & U_m \end{pmatrix},$$

see Sect. 7.2.

The vector $\boldsymbol{\vartheta}$ consists of both σ^2 and the elements of $\boldsymbol{Q}$, without more specific assumptions. The noninformative (but improper) Jeffreys' prior

$$p(\sigma^2) \propto \sigma^{-2}, \quad p(\boldsymbol{Q}) \propto |\boldsymbol{Q}|^{-\frac{q+2}{2}},$$

with $q+1 = \dim(\boldsymbol{y}_i)$, can lead to improper posteriors such that MCMC techniques also break down. Therefore, a weakly informative inverse gamma prior $\sigma^2 \sim \text{IG}(a, b)$ is commonly proposed with small values a and b . Note that meaningful values for a and b depend on the magnitude of the response observations. We have good experience with $a = b = 0.001$ if the responses are divided by their standard deviation prior to estimation. For $\boldsymbol{Q}$, an inverse Wishart distribution may be specified as a prior, generalizing an inverse gamma distribution to the multivariate case. However, our experience shows, that a full Wishart prior works in practice only for two or three random effects (if at all). To simplify, we further assume that $\boldsymbol{Q}$ is a diagonal matrix with elements τ_r^2 , $r = 0, \dots, q$. The inverse Wishart distribution then factorizes into the product of inverse gamma priors

$$\tau_r^2 \sim \text{IG}(a_r, b_r), \quad r = 0, \dots, q.$$

We therefore implicitly assume that the components γ_{ir} in $\boldsymbol{y}_i$ are conditionally independent a priori with priors

$$\gamma_{ir} | \tau_r^2 \sim \text{N}(0, \tau_r^2), \quad r = 0, \dots, q.$$

The full conditionals then become $\sigma^2 | \cdot \sim \text{IG}(\tilde{a}, \tilde{b})$ and $\tau_r^2 | \cdot \sim \text{IG}(\tilde{a}_r, \tilde{b}_r)$, $r = 0, \dots, q$, with hyperparameters provided in Box 7.4. Note that the full conditional for $\boldsymbol{y}$ in the general LMM considerably simplifies due to the independence assumptions on the γ_{ir} . In fact, the γ_{ir} can be updated independently by sampling from univariate normal densities. The resulting Gibbs sampler is provided in Box 7.4. Note that this sampler is derived under special prior assumptions, in particular independent inverse gamma priors, for the variances of the random effects priors. The latter assumption is sometimes criticized, and alternatives are suggested; see Gelman (2006). When using inverse gamma priors it is important to investigate how sensitive results are with respect to the values a_r and b_r of the hyperparameters; see the following Example 7.4.

7.4 Gibbs Sampler for Longitudinal and Clustered Data

Model

$$y_{ij} | \boldsymbol{\beta}, \boldsymbol{\gamma}_i, \sigma^2 \sim N(\mathbf{x}'_{ij}\boldsymbol{\beta} + \mathbf{u}'_{ij}\boldsymbol{\gamma}_i, \sigma^2), \quad i = 1, \dots, m, \quad j = 1, \dots, n_i.$$

Priors

- $\boldsymbol{\beta} \sim N(\mathbf{m}, \mathbf{M})$, where $\mathbf{m}$ and $\mathbf{M}$ are assumed to be known.
- The random effects parameters $\boldsymbol{\gamma}_{ir}$ are conditionally independent with $\boldsymbol{\gamma}_{ir} | \tau_r^2 \sim N(0, \tau_r^2)$, $i = 1, \dots, m$, $r = 0, \dots, q$.
- The variance parameters are independent inverse gamma distributed, i.e., $\sigma^2 \sim \text{IG}(a, b)$ and $\tau_r^2 \sim \text{IG}(a_r, b_r)$, $r = 0, \dots, q$.

Gibbs Sampler

1. Initialization: Specify initial values $\boldsymbol{\beta}^{(0)}, \boldsymbol{\gamma}_i^{(0)}$, $i = 1, \dots, m$, $(\sigma^2)^{(0)}$, $(\tau_r^2)^{(0)}$, $r = 0, \dots, q$ and the number of iterations T . Set $t = 1$.
2. Sample $\boldsymbol{\beta}^{(t)}$ from $N(\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta)$ with

$$\boldsymbol{\Sigma}_\beta = \left(\frac{1}{\sigma^2} \mathbf{X}'\mathbf{X} + \mathbf{M}^{-1} \right)^{-1}, \quad \boldsymbol{\mu}_\beta = \boldsymbol{\Sigma}_\beta (\mathbf{M}^{-1}\mathbf{m} + \frac{1}{\sigma^2} \mathbf{X}'(\mathbf{y} - \mathbf{U}\boldsymbol{\gamma})).$$

Use $\boldsymbol{\gamma} = \boldsymbol{\gamma}^{(t-1)}$, $\sigma^2 = (\sigma^2)^{(t-1)}$ in $\boldsymbol{\mu}_\beta$ and $\boldsymbol{\Sigma}_\beta$.

3. For $r = 0, \dots, q$ sample $\boldsymbol{\gamma}_{ir}^{(t)}$ from $N(\boldsymbol{\mu}_{\boldsymbol{\gamma}_{ir}}, \sigma_{\boldsymbol{\gamma}_{ir}}^2)$ with

$$\sigma_{\boldsymbol{\gamma}_{ir}}^2 = \sigma^2 \left(\sum_{j=1}^{n_i} u_{ijr}^2 + \sigma^2 / \tau_r^2 \right)^{-1}, \quad \boldsymbol{\mu}_{\boldsymbol{\gamma}_{ir}} = \frac{\sigma_{\boldsymbol{\gamma}_{ir}}^2}{\sigma^2} \sum_{j=1}^{n_i} u_{ijr} (y_{ij} - \mathbf{x}'_{ij}\boldsymbol{\beta}).$$

Use $\boldsymbol{\beta} = \boldsymbol{\beta}^{(t)}$, $\sigma^2 = (\sigma^2)^{(t-1)}$, $(\tau_r^2)^{(t-1)}$ in $\boldsymbol{\mu}_{\boldsymbol{\gamma}_{ir}}$ and $\sigma_{\boldsymbol{\gamma}_{ir}}^2$.

4. Sample $(\sigma^2)^{(t)}$ from $\text{IG}(\tilde{a}, \tilde{b})$ with

$$\tilde{a} = a + \frac{1}{2}, \quad \tilde{b} = b + \frac{1}{2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{U}\boldsymbol{\gamma})' (\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{U}\boldsymbol{\gamma}).$$

Use $\boldsymbol{\beta} = \boldsymbol{\beta}^{(t)}$, $\boldsymbol{\gamma} = \boldsymbol{\gamma}^{(t)}$ in $\tilde{a}$ and $\tilde{b}$.

5. For $r = 0, \dots, q$ sample $(\tau_r^2)^{(t)}$ from $\text{IG}(\tilde{a}_r, \tilde{b}_r)$ with

$$\tilde{a}_r = a_r + \frac{m}{2}, \quad \tilde{b}_r = b_r + \frac{1}{2} \sum_{i=1}^m \boldsymbol{\gamma}_{ir}^2.$$

Use $\boldsymbol{\gamma}_{ir} = \boldsymbol{\gamma}_{ir}^{(t)}$ in $\tilde{a}_r$ and $\tilde{b}_r$.

6. Stop if $t = T$, otherwise set $t = t + 1$ and go to step 2.

Table 7.3 Hormone therapy with rats: estimated fixed effects in a fully Bayesian approach

Variable	Coefficient	Standard deviation	2.5 % quantile	97.5 % quantile
<i>intercept</i>	68.610	0.343	67.895	69.272
<i>low-dose</i>	7.500	0.227	7.066	7.953
<i>high-dose</i>	6.862	0.235	6.402	7.307
<i>control</i>	7.318	0.284	6.764	7.902

Example 7.4 Hormone Therapy with Rats—Bayesian Linear Mixed Model

We reanalyze the random intercept model (7.13) in Example 7.1, specifying inverse gamma priors for σ^2 and $\tau_0^2 = \text{Var}(\gamma_{0i})$ with $a = a_0 = b = b_0 = 0.001$. Table 7.3 shows estimates corresponding to Table 7.1 and, additionally, 2.5 %- and 97.5 %-quantiles. The results are obtained with the software package BayesX and are based on 102,000 iterations and using every 100th sampled parameter for estimation after the burn in period of 2,000 iterations. The estimates are in good agreement with corresponding ML estimates. The posterior mean of the random effects variance is 3.674 and hence also close to the likelihood-based estimator. A sensitivity analysis regarding the choice of the hyperparameters a_0 and b_0 with alternative choices $a_0 = b_0 = 0.01$ and $a_0 = b_0 = 0.0001$ showed no substantially different results.

△

7.5 Generalized Linear Mixed Models

Similar to linear models, the linear predictor in generalized linear models will be extended by including random effects. This leads to GLMMs, generalizing LMMs to models for non-Gaussian responses that are, for example, binary, discrete, or nonnegative. As expected, statistical inference becomes technically more involved when moving from LMMs to GLMMs.

7.5.1 GLMMs for Longitudinal and Clustered Data

Recall the definitions in Sect. 5.5 (p. 304). The (conditional) density of y_i , given the linear predictor $\eta_i = \mathbf{x}_i' \boldsymbol{\beta}$, belongs to an exponential family (e.g., has a binomial, Poisson, or gamma distribution) and the linear predictor is related to the mean $\mu_i = E(y_i)$ through

$$\mu_i = h(\eta_i) \quad \text{and} \quad \eta_i = g(\mu_i).$$

To define GLMMs, we extend the linear predictor $\eta_i = \mathbf{x}_i' \boldsymbol{\beta}$ of a GLM by adding random effects.

The responses y_{ij} , $i = 1, \dots, m$, $j = 1, \dots, n_i$, with n_i repeated measurements per individual or cluster, may now be, for example, binary, or count variables. Including individual- or cluster-specific random effects $\boldsymbol{\gamma}_i$, responses y_{ij} given $\boldsymbol{\gamma}_i$ are then (conditionally) independent and could be, e.g., binomial or Poisson

distributed. The conditional mean $\mu_{ij} = E(y_{ij} | \mathbf{y}_i)$ is related to a linear mixed predictor of the same form as for the LMM, i.e.,

$$\eta_{ij} = \mathbf{x}'_{ij}\boldsymbol{\beta} + \mathbf{u}'_{ij}\boldsymbol{\gamma}_i, \quad i = 1, \dots, m, \quad j = 1, \dots, n_i,$$

through the relation $\mu_{ij} = h(\eta_{ij})$, with a suitable response function h .

As in the LMM, $\mathbf{u}_{ij}$ is usually a subvector of $\mathbf{x}_{ij}$. In particular, the choice $\mathbf{u}_{ij} \equiv 1$ defines a random intercept model

$$\eta_{ij} = \mathbf{x}'_{ij}\boldsymbol{\beta} + \gamma_{0i} = \beta_0 + \beta_1 x_{ij1} + \dots + \beta_k x_{ijk} + \gamma_{0i},$$

where γ_{0i} are random deviations from the fixed intercept β_0 . For random effects, we make the same assumptions as for LMMs, i.e., they are independent and identically normal with $\boldsymbol{\gamma}_i \sim N(\mathbf{0}, \boldsymbol{Q})$ and independent from covariates.

The assumption of conditionally independent responses, given the random effects, corresponds to the assumption of independent errors in LMMs. This assumption may be relaxed by assuming appropriate dependency structures, e.g., for odds ratios with binary responses. However, such generalized approaches in GLMMs are typically much more complicated compared to introducing correlated residuals in LMMs.

A summary of GLMMs for longitudinal and clustered data is given in Box 7.5. We will now discuss some specific GLMMs.

Mixed Logit and Probit Models

For binary response variables, we obtain mixed logit models, for example,

$$\log \frac{P(y_{ij} = 1 | \boldsymbol{\gamma}_i)}{P(y_{ij} = 0 | \boldsymbol{\gamma}_i)} = \mathbf{x}'_{ij}\boldsymbol{\beta} + \mathbf{u}'_{ij}\boldsymbol{\gamma}_i. \quad (7.41)$$

As in Chap. 5, mixed probit models are a possible alternative.

Example 7.5 Speed Dating—Mixed Logit Model

We will illustrate (binary) mixed logit models with data from [Fisman et al. \(2006\)](#) on a speed dating experiment on a sample of a few hundred students in graduate and professional schools at Columbia University. In the speed dating events, the experiment randomly assigned each participant to ten short dates with participants of the opposite sex. For each date, each person rated six attributes (attractive, sincere, intelligent, fun, ambitious, shared interests) of the other person on a 10-point scale and wrote down whether he or she would like to see the other person again. We denote these repeated binary decisions as

$$y_{ij} = \begin{cases} 1 & \text{if person } i \text{ likes to see the } j \text{th partner again} \\ 0 & \text{otherwise,} \end{cases}$$

and analyze the data with a mixed logit model of the form (7.41), where the covariates consist of the ratings, as well as the gender of the person. A similar example, also based on the speed dating data, appears in the exercises to Chap. 14 in [Gelman and Hill \(2006\)](#). $\triangle$

7.5 GLMMs for Longitudinal and Clustered Data

Distributional Assumption

Given the random effects $\boldsymbol{\gamma}_i$ and the covariates $\mathbf{x}_{ij}$, $\mathbf{u}_{ij}$, the responses y_{ij} are conditionally independent and the conditional density $f(y_{ij} | \boldsymbol{\gamma}_i)$ belongs to an exponential family as in GLMs; see Sect. 5.5 (p. 304).

Structural Assumption

The conditional mean $\mu_{ij} = E(y_{ij} | \boldsymbol{\gamma}_i)$ is linked to the linear predictor

$$\eta_{ij} = \mathbf{x}_{ij}'\boldsymbol{\beta} + \mathbf{u}_{ij}'\boldsymbol{\gamma}_i$$

through

$$\mu_{ij} = h(\eta_{ij}) \quad \text{or} \quad \eta_{ij} = g(\mu_{ij}),$$

where h is the response function and $g = h^{-1}$ the link function.

Distributional Assumption for Random Effects

The random effects $\boldsymbol{\gamma}_i$, $i = 1, \dots, m$, are independent and identically distributed with

$$\boldsymbol{\gamma}_i \stackrel{i.i.d.}{\sim} N(\mathbf{0}, \boldsymbol{Q})$$

and positive definite covariance matrix $\boldsymbol{Q}$. An important special case is $\boldsymbol{Q} = \text{diag}(\tau_0^2, \dots, \tau_q^2)$.

Software

- Function `gllamm` of STATA; see [Skron dal and Rabe-Hesketh \(2008\)](#)
- Package `lme4` of R
- `proc mixed` of SAS
- Software package BayesX (also see the R interface R2BayesX)

Mixed Poisson Models

For count responses $y_{ij} \sim \text{Po}(\lambda_{ij} | \boldsymbol{\gamma}_i)$ mixed log-linear Poisson models

$$\log(\lambda_{ij}) = \mathbf{x}_{ij}'\boldsymbol{\beta} + \mathbf{u}_{ij}'\boldsymbol{\gamma}_i,$$

are the most common choice.

While repeated measurements ($n_i > 1$) are usually needed to identify binary random intercept models, this is not necessary for mixed Poisson models. It is possible to specify the (random) rate without repeated observations, i.e.,

$$\lambda_i = \exp(\mathbf{x}_i' \boldsymbol{\beta} + \gamma_{0i}), \quad (7.42)$$

with $\gamma_{0i} \sim N(0, \tau_0^2)$ in a random intercept model. A log-linear random intercept model is also called a Poisson-normal or Poisson-lognormal model because Eq. (7.42) is equivalent to

$$\lambda_i = v_i \exp(\mathbf{x}_i' \boldsymbol{\beta}),$$

where $v_i = \exp(\gamma_{0i})$ has a lognormal distribution. Poisson-(log)normal models can be used to analyze count data with overdispersion; see the following example and the discussion on the implied marginal model in Sect. 7.5.2.

Example 7.6 Number of Citations of Patents—Mixed Poisson Regression

In Example 5.7 (p. 297), log-linear Poisson models, both with and without overdispersion, have been employed for regression analyzes using the response *ncit* (number of citations). There is clear evidence of overdispersion, and log-linear Poisson random intercept models (7.42) provide an alternative for such modeling. For illustrative purposes, we consider a log-linear model with linear effects and extend it to the log-linear Poisson-normal model

$$\begin{aligned} \log(\lambda_i) = \eta_i = & \beta_0 + \beta_1 \text{yearc}_i + \beta_2 \text{ncountryc}_i + \beta_3 \text{nclaimsc}_i + \beta_4 \text{biopharm}_i \\ & + \beta_5 \text{sustwin}_i + \beta_6 \text{patus}_i + \beta_7 \text{patgsgr}_i + \beta_8 \text{opp}_i + \gamma_{0i}, \end{aligned}$$

where again *yearc*, *ncountryc*, and *nclaimsc* are the centered versions of the covariates. Δ

Mixed Models for Categorical Responses

Mixed models can also be formulated for categorical responses through suitable generalization of predictors in categorical regression models (Chap. 6). For example, multinomial logit mixed models for a response Y with unordered categories $(1, \dots, c + 1)$ that include global covariates, but additionally include category-specific fixed effects $\boldsymbol{\beta}_r$ and random effects $\boldsymbol{\gamma}_r$, are defined through

$$P(Y_{ij} = r \mid \boldsymbol{\gamma}_r) = \frac{\exp(\eta_{ijr})}{1 + \sum_{s=1}^c \exp(\eta_{ijs})} \quad r = 1, \dots, c,$$

with category-specific predictors

$$\eta_{ijr} = \mathbf{x}_{ij}' \boldsymbol{\beta}_r + \mathbf{u}_{ij}' \boldsymbol{\gamma}_{ir}.$$

The random effects $\boldsymbol{\gamma}_{i1}, \dots, \boldsymbol{\gamma}_{i,c}$ are assumed to be i.i.d. Gaussian, $\boldsymbol{\gamma}_{ir} \sim N(\mathbf{0}, \boldsymbol{Q}_r)$.

7.5.2 Conditional and Marginal Models

Conceptually, the marginal density $p(y_{ij})$ can be obtained by integrating the conditional density $p(y_{ij} \mid \boldsymbol{\gamma}_i)$ determined through the specific exponential family, i.e.,

$$p(y_{ij}) = \int p(y_{ij} | \boldsymbol{\gamma}_i) p(\boldsymbol{\gamma}_i) d\boldsymbol{\gamma}_i.$$

The density $p(\boldsymbol{\gamma}_i)$ of $\boldsymbol{\gamma}_i$ is a $N(\mathbf{0}, \mathbf{Q})$ distribution. Apart from the case of LMMs, where $p(y_{ij} | \boldsymbol{\gamma}_i)$ also follows a normal distribution, the integration only can be carried out analytically in rather special cases. This is also one of the main reasons why statistical inference in GLMMs becomes more complex compared to LMMs. Marginal means, variances, and covariances are also only available analytically in special cases.

For the log-linear Poisson random intercept model (7.42) with $n_i = 1$, the marginal means, variances, and covariances can all be determined analytically because

$$\mu_i | \gamma_{0i} = v_i \exp(\mathbf{x}'_i \boldsymbol{\beta}),$$

and $v_i = \exp(\gamma_{0i})$ has a lognormal distribution. Thus

$$E(y_i) = E(v_i \exp(\mathbf{x}'_i \boldsymbol{\beta})) = \exp(\tau_0^2/2) \exp(\mathbf{x}'_i \boldsymbol{\beta}) = \exp(\tau_0^2/2 + \mathbf{x}'_i \boldsymbol{\beta}).$$

Up to the intercept term, the fixed population effects $\boldsymbol{\beta}$ are identical in the marginal and in the conditional model. Furthermore, it can be shown that

$$\text{Var}(y_i) = E(y_i) \{1 + \exp(\mathbf{x}'_i \boldsymbol{\beta}) (\exp(3\tau_0^2/2) - \exp(\tau_0^2/2))\}.$$

Since the second factor is greater than 1, the marginal variance is larger than the marginal mean. Therefore the marginal distribution is not a Poisson distribution, but rather a distribution with overdispersion. It is not possible, however, to derive an explicit form for its density. In addition, it can be shown that

$$\text{Cov}(y_i, y_j) = \exp(\mathbf{x}'_i \boldsymbol{\beta} + \mathbf{x}'_j \boldsymbol{\beta}) \{ \exp(\tau_0^2) (\exp(\tau_0^2) - 1) \}.$$

Therefore, y_i and y_j have positive marginal correlations, with specific covariance structure.

In contrast to the log-linear Poisson model, particularly in contrast to the LMM, fixed covariate effects in conditional and marginal models are generally different. For example, in the case of a probit-normal random intercept model

$$P(y_{ij} | \gamma_{0i}) = P(y_{ij} = 1 | \gamma_{0i}) = \Phi(\mathbf{x}'_{ij} \boldsymbol{\beta} + \gamma_{0i}), \quad \gamma_{0i} \sim N(0, \tau_0^2),$$

it can be shown that

$$P(y_{ij} = 1) = \Phi\left(\frac{\mathbf{x}'_{ij} \boldsymbol{\beta}}{\sqrt{\tau_0^2 + 1}}\right) = \Phi(\mathbf{x}'_{ij} \boldsymbol{\beta}^*),$$

with $\beta^* = \beta / \sqrt{\tau_0^2 + 1}$. Thus absolute values of fixed effects in the marginal model are shrunk by the factor $1 / \sqrt{\tau_0^2 + 1}$. For a logit model with the same predictor, it is not even possible to derive any explicit formula, rather only an approximation based on Taylor expansion; see, for example, [Fahrmeir and Tutz \(2001, Sect. 7.7\)](#).

7.5.3 GLMMs in General Form

Similar to the LMMs in Sect. 7.3, the predictor vector η for all observations can be expressed as

$$\eta = X\beta + U\gamma,$$

where X and U are appropriate design matrices and $\gamma = (\gamma'_1, \dots, \gamma'_m)'$ is the vector of all random effects. As for the LMM, we may replace the assumption

$$\gamma_i \stackrel{i.i.d.}{\sim} N(0, Q),$$

more generally with $\gamma \sim N(0, G)$, where G is any positive definite covariance matrix. This allows for correlated random effects, as needed in the mixed model representation of spline functions in Chaps. 8 and 9. Furthermore, statistical inference in the next section is formulated using this compact notation for the distribution of γ .

For GLMMs in general form, we allow X and U to be general design matrices and G may be any positive definite covariance matrix. This is completely analogous to general LMMs in Sect. 7.2. In the following section on inference we will refer to a specific observation in a general GLMM by the index i , i.e., y_i denotes the i th observation.

7.6 Likelihood and Bayesian Inference in GLMMs

Conceptually, inference in GLMMs is based on the same likelihood or Bayes principles as in LMMs. However, since the conditional likelihood is in general non-Gaussian and the relationship between mean and predictor is nonlinear in GLMMs, important parts of these approaches cannot be carried out analytically but rather have to be replaced by numerical methods or suitable approximations. We focus on an approximate penalized likelihood approach and on MCMC techniques for fully Bayesian inference. Both approaches are strongly related since the penalized likelihood estimator can also be seen as an empirical Bayes estimator. For simplicity, we restrict the presentation to models with scale parameter $\phi = 1$; however, the more general case, such as for gamma-distributed responses, can be dealt with as in the GLM.

7.6.1 Penalized Likelihood and Empirical Bayes Estimation

Known Variance–Covariance Parameters

We first assume that all parameters in $\mathbf{G}$ are known.

Similar to Sect. 7.3.1, we consider the joint likelihood

$$L(\boldsymbol{\beta}, \boldsymbol{\gamma}) = p(\mathbf{y} \mid \boldsymbol{\beta}, \boldsymbol{\gamma}) p(\boldsymbol{\gamma})$$

for known $\mathbf{G}$, and maximize it simultaneously with respect to $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$. Since

$$p(\boldsymbol{\beta}, \boldsymbol{\gamma} \mid \mathbf{y}) \propto p(\mathbf{y} \mid \boldsymbol{\beta}, \boldsymbol{\gamma}) p(\boldsymbol{\gamma}),$$

this is equivalent to determining the posterior mode estimator $(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\gamma}})$ or an *empirical Bayes estimator*. Taking logarithms, we obtain the penalized log-likelihood

$$l_{\text{pen}}(\boldsymbol{\beta}, \boldsymbol{\gamma}) = l(\boldsymbol{\beta}, \boldsymbol{\gamma}) - \frac{1}{2} \boldsymbol{\gamma}' \mathbf{G}^{-1} \boldsymbol{\gamma}. \quad (7.43)$$

The log-likelihood $l(\boldsymbol{\beta}, \boldsymbol{\gamma})$ is defined as in a GLM, where only the predictor $\eta_i = \mathbf{x}_i' \boldsymbol{\beta}$ for a particular observation i has to be replaced by the extended mixed model predictor $\eta_i = \mathbf{x}_i' \boldsymbol{\beta} + \mathbf{u}_i' \boldsymbol{\gamma}$. Maximization of $l_{\text{pen}}(\boldsymbol{\beta}, \boldsymbol{\gamma})$ proceeds in complete analogy to simple GLMs as outlined in Sect. 5.4.2 and fully derived in Sect. 5.8.2. With $\mathbf{C} = (\mathbf{X} \ \mathbf{U})$ and the partitioned matrix

$$\mathbf{B} = \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{G}^{-1} \end{pmatrix},$$

the resulting iteratively weighted least squares estimator

$$\begin{pmatrix} \hat{\boldsymbol{\beta}}^{(t+1)} \\ \hat{\boldsymbol{\gamma}}^{(t+1)} \end{pmatrix} = (\mathbf{C}' \mathbf{W}^{(t)} \mathbf{C} + \mathbf{B})^{-1} \mathbf{C}' \mathbf{W}^{(t)} \tilde{\mathbf{y}}^{(t)}$$

has the same form as the ML or penalized least squares estimator in the LMM; see formula (7.25) of Sect. 7.3.1. The matrix of working weights $\mathbf{W}^{(t)}$ and the working observations $\tilde{\mathbf{y}}_i$ are defined in complete analogy to simple GLMs. More specifically, we obtain for a particular observation $\mathbf{x}_i, \mathbf{u}_i$ in the general GLMM

$$\tilde{y}_i \left(\hat{\eta}_i^{(t)} \right) = \hat{\eta}_i^{(t)} + \frac{\left(y_i - h \left(\hat{\eta}_i^{(t)} \right) \right)}{h' \left(\hat{\eta}_i^{(t)} \right)}, \quad (7.44)$$

where $\hat{\eta}_i^{(t)} = \mathbf{x}_i' \hat{\boldsymbol{\beta}}^{(t)} + \mathbf{u}_i' \hat{\boldsymbol{\gamma}}^{(t)}$ is the actual predictor, h is the response function, and $h'(\eta) = \partial h(\eta) / \partial \eta$ is the derivative of h with respect to η . The working weight is given by

$$\tilde{w}_i \left(\hat{\eta}_i^{(t)} \right) = \frac{\left(h' \left(\hat{\eta}_i^{(t)} \right) \right)^2}{\sigma_i^2 \left(\hat{\eta}_i^{(t)} \right)}, \quad (7.45)$$

where $\sigma_i^2 \left(\hat{\eta}_i^{(t)} \right)$ is the (conditional) variance $\text{Var}(y_i)$ evaluated at $\eta = \hat{\eta}_i^{(t)}$. Omitting the iteration index we obtain in matrix notation

$$\begin{aligned} \mathbf{W} &= \mathbf{D} \mathbf{\Sigma}^{-1} \mathbf{D}, \\ \tilde{\mathbf{y}} &= \mathbf{X} \hat{\boldsymbol{\beta}} + \mathbf{U} \hat{\boldsymbol{\gamma}} + \mathbf{D}^{-1} (\mathbf{y} - \boldsymbol{\mu}), \end{aligned} \quad (7.46)$$

with $\mathbf{D} = \text{diag} \left(\dots, \frac{\partial h(\eta_i)}{\partial \eta}, \dots \right)$ and $\mathbf{\Sigma} = \text{diag}(\dots, \sigma_i^2, \dots)$.

Unknown Variance–Covariance Parameters

Now assume that the covariance matrix $\mathbf{G} = \mathbf{G}(\boldsymbol{\vartheta})$ contains unknown parameters $\boldsymbol{\vartheta}$. Estimation of the variance–covariance parameters $\boldsymbol{\vartheta}$ can be accomplished by means of an approximate marginal likelihood. A Laplace approximation argument (Breslow and Clayton, 1993) shows that the conditional log-likelihood can be approximated through a quadratic form:

$$l(\boldsymbol{\beta}, \boldsymbol{\gamma}; \boldsymbol{\vartheta}) = \log p(\mathbf{y} | \boldsymbol{\beta}, \boldsymbol{\gamma}) \approx \frac{1}{2} (\mathbf{y} - \boldsymbol{\mu})' \mathbf{\Sigma}^{-1} (\mathbf{y} - \boldsymbol{\mu}),$$

where $\boldsymbol{\mu}$ and $\mathbf{\Sigma}$ depend on $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$. From the definition Eq. (7.46) of the working observations $\tilde{\mathbf{y}}$, we get

$$\mathbf{y} - \boldsymbol{\mu} = \mathbf{D} (\tilde{\mathbf{y}} - \mathbf{X} \hat{\boldsymbol{\beta}} - \mathbf{U} \hat{\boldsymbol{\gamma}}),$$

which implies (together with $\mathbf{W} = \mathbf{D} \mathbf{\Sigma}^{-1} \mathbf{D}$)

$$l(\boldsymbol{\beta}, \boldsymbol{\gamma}; \boldsymbol{\vartheta}) \approx (\tilde{\mathbf{y}} - \mathbf{X} \hat{\boldsymbol{\beta}} - \mathbf{U} \hat{\boldsymbol{\gamma}})' \mathbf{W} (\tilde{\mathbf{y}} - \mathbf{X} \hat{\boldsymbol{\beta}} - \mathbf{U} \hat{\boldsymbol{\gamma}}).$$

Therefore, the conditional log-likelihood of a GLMM can be approximated for fixed $\boldsymbol{\vartheta}$ through the log-likelihood of the LMM

$$\tilde{\mathbf{y}} | \boldsymbol{\gamma} \sim \text{N}(\mathbf{X} \boldsymbol{\beta} + \mathbf{U} \boldsymbol{\gamma}, \mathbf{W}^{-1}).$$

Setting

$$\mathbf{V} = \mathbf{W} + \mathbf{U} \mathbf{G}(\boldsymbol{\vartheta})^{-1} \mathbf{U}',$$

we obtain the (approximate) *marginal* or *restricted* log-likelihood

$$l_R(\boldsymbol{\vartheta}) = -\frac{1}{2} \log |\mathbf{V}(\boldsymbol{\vartheta})| - \frac{1}{2} \log (|\mathbf{X}' \mathbf{V}(\boldsymbol{\vartheta})^{-1} \mathbf{X}|) - \frac{1}{2} (\tilde{\mathbf{y}}(\boldsymbol{\vartheta}) - \mathbf{X} \hat{\boldsymbol{\beta}})' \mathbf{V}(\boldsymbol{\vartheta})^{-1} (\tilde{\mathbf{y}}(\boldsymbol{\vartheta}) - \mathbf{X} \hat{\boldsymbol{\beta}}),$$

where $\mathbf{V}(\boldsymbol{\vartheta})$ and $\tilde{\mathbf{y}}(\boldsymbol{\vartheta})$ also depend on $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\gamma}}$.

This leads to the estimation algorithm in Box 7.6, which switches between estimation of β and γ and estimation of ϑ .

The estimated covariance matrix is the inverse of the estimated Fisher information matrix $\hat{F} = F(\hat{\beta}, \hat{\gamma}, \hat{\vartheta})$, i.e.,

$$\widehat{\text{Cov}} \begin{pmatrix} \hat{\beta} \\ \hat{\gamma} \end{pmatrix} \approx \hat{F}^{-1}.$$

Assuming approximate normality, confidence intervals and tests of linear hypotheses can be constructed analogously to those for the LMM. If the number of parameters is large relative to the sample size, we expect increased bias and a degraded normality approximation, in particular for responses with low information, such as binary observations. Additional analyses that use fully Bayesian inference are recommended in such cases.

7.6.2 Fully Bayesian Inference Using MCMC

For fully Bayesian inference, we assume the same priors for β and ϑ as in LMMs, i.e.,

$$\beta \sim N(m, M),$$

with the limiting case $M^{-1} \rightarrow \mathbf{0}$ corresponding to a flat prior

$$p(\beta) \propto \text{const}.$$

Usually Bayesian inference using MCMC requires knowledge of the specific structure of the GLMM. Bayesian inference for a general GLMM is usually difficult to accomplish, and there is no guarantee that the sampler works.

We therefore restrict ourselves to the specific GLMM for longitudinal or clustered data,

$$E(y_{ij} | \gamma_i) = \mu_{ij} = h(x'_{ij}\beta + u'_{ij}\gamma_i), \quad i = 1, \dots, m, \quad j = 1, \dots, n_i,$$

with i.i.d. random effects

$$\gamma_i \sim N(\mathbf{0}, Q).$$

We also make the simplifying assumption $Q = \text{diag}(\tau_0^2, \dots, \tau_q^2)$ and assume a priori independent diagonal elements with

$$\tau_r^2 \sim \text{IG}(a_r, b_r), \quad r = 0, \dots, q.$$

Moreover, we only consider GLMMs with scale parameter $\phi = 1$, otherwise an additional prior for ϕ has to be specified.

7.6 Penalized Likelihood/Empirical Bayes Estimation in GLMMs

Score Function and Fisher Information

$$s_{\beta}(\beta, \gamma, \vartheta) = \frac{\partial l_{\text{pen}}(\beta, \gamma; \vartheta)}{\partial \beta} = X' D \Sigma^{-1} (y - \mu),$$

$$s_{\gamma}(\beta, \gamma, \vartheta) = \frac{\partial l_{\text{pen}}(\beta, \gamma; \vartheta)}{\partial \gamma} = U' D \Sigma^{-1} (y - \mu) - G(\vartheta)^{-1} \gamma.$$

$$F(\beta, \gamma, \vartheta) = \begin{pmatrix} X' W X & X' W U \\ U' W X & U' W U + G(\vartheta)^{-1} \end{pmatrix}$$

with $D = \text{diag}(\dots, \frac{\partial h(\eta_i)}{\partial \eta}, \dots)$, $\Sigma = \text{diag}(\dots, \sigma_i^2, \dots)$, and $\mu = (\dots, h(\eta_i), \dots)'$.

Numerical Computation

1. Initialize with starting values $\hat{\beta}^{(0)}, \hat{\gamma}^{(0)}, \hat{\vartheta}^{(0)}$. Set $t = 1$.
2. Determine $\hat{\beta}^{(t+1)}$ and $\hat{\gamma}^{(t+1)}$ through

$$\begin{pmatrix} \hat{\beta}^{(t+1)} \\ \hat{\gamma}^{(t+1)} \end{pmatrix} = (C' W^{(t)} C + B)^{-1} C' W^{(t)} \tilde{y}^{(t)}$$

with working weights and observations as given in Eqs. (7.45) and (7.44).

3. Compute the (approximate) ML estimator $\hat{\vartheta}^{(t+1)}$ through one iteration step of some numerical algorithm for maximizing $l_R(\vartheta)$, for example, a Newton–Raphson step.
4. Terminate the algorithm if convergence (according to some stopping criterion) is reached and obtain $\hat{\beta}, \hat{\gamma}, \hat{\vartheta}$. Otherwise, set $t = t + 1$ and proceed with 2.

Based on this assumptions, we obtain the posterior

$$p(\beta, \gamma_1, \dots, \gamma_m, Q | y) \propto \prod_{i=1}^m \prod_{j=1}^{n_i} p(y_{ij} | \beta, \gamma_i) p(\beta) \prod_{i=1}^m p(\gamma_i | Q) p(Q).$$

This implies the *full conditionals*

$$\begin{aligned}
p(\boldsymbol{\beta} | \cdot) &\propto \prod_{i=1}^m \prod_{j=1}^{n_i} p(y_{ij} | \boldsymbol{\beta}, \boldsymbol{\gamma}_i) p(\boldsymbol{\beta}) \\
p(\boldsymbol{\gamma}_i | \cdot) &\propto \prod_{j=1}^{n_i} p(y_{ij} | \boldsymbol{\beta}, \boldsymbol{\gamma}_i) p(\boldsymbol{\gamma}_i | \boldsymbol{Q}), \quad i = 1, \dots, m \\
p(\tau_r^2 | \cdot) &\propto \prod_{i=1}^m p(\boldsymbol{\gamma}_{ir} | \tau_r^2) p(\tau_r^2), \quad r = 0, \dots, q.
\end{aligned}$$

In contrast to the LMM, full conditionals for $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}_i$, $i = 1, \dots, m$, are no longer available in a known analytical form. Therefore random numbers cannot be directly drawn from these full conditionals, and thus the Metropolis–Hastings (MH) algorithm needs to be employed instead of the Gibbs sampler (as for LMMs in Sect. 7.4). We prefer the automated algorithm of [Gamerman \(1997\)](#) containing IWLS (iteratively weighted least squares) proposals. See Sect. 5.6.2 for the general idea of IWLS proposals.

The full conditionals $p(\tau_r^2 | \cdot)$ are again inverse gamma densities

$$\tau_r^2 | \cdot \sim \text{IG}(\tilde{a}_r, \tilde{b}_r)$$

with updated parameters

$$\tilde{a}_r = a_r + \frac{m}{2}, \quad \tilde{b}_r = b_r + \frac{1}{2} \sum_{i=1}^m \gamma_{ir}^2.$$

Alternatively the data augmentation schemes outlined in Sect. 5.6.3 for GLMs could be applied in a straightforward way for GLMMs.

Example 7.7 Speed Dating—Mixed Binary Regression

To analyze binary repeated measurement data from the speed dating experiment (Example 7.5), we first reduced the sample by deleting all participants who never wanted or always wanted to see a partner again. Without these “outliers,” $m = 390$ persons remained in the sample. For illustration, we only included the two attributes “attractive” and “shared interests,” as well as gender, and the interactions between gender and the two attributes as fixed effects into the predictor η_{ij} of a mixed logit model for $P(Y_{ij} = 1 | \eta_{ij})$, the probability that person i likes to see partner j again. For a random intercept model, the predictor is

$$\begin{aligned}
\eta_{ij} = & \beta_0 + \beta_1 \text{gender}_i + \beta_2 \text{attr}_{ij} + \beta_3 \text{shar}_{ij} \\
& + \beta_4 \text{gender}_i \cdot \text{attr}_{ij} + \beta_5 \text{gender}_{ij} \cdot \text{shar}_{ij} + \gamma_{0i}.
\end{aligned}$$

Gender is dummy-coded (male = 1, female = 0) and the two attributes are rated on a 10-point scale. The variables *attr* and *shar* are standardized prior to estimation. With such a model we can analyze the effect of gender and the main effects of attractiveness and shared interests on the decision to see a partner again. The interaction terms allow analysis as to whether or not there are gender-specific differences of the effects of the two attributes. The intercept term γ_{0i} captures omitted covariates and induces correlation between repeated



Table 7.4 Speed dating: random intercept and random slope models based on likelihood (L) and Bayesian (B) inference. Posterior estimates and standard errors for fixed effects, (posterior) variance estimates for random effects

Covariate	L: random intercept		L: random slopes	
<i>intercept</i>	−0.6976	(0.118)	−0.7997	(0.126)
<i>gender</i>	−0.1751	(0.166)	0.2823	(0.177)
<i>attr</i>	1.6329	(0.077)	1.9323	(0.105)
<i>shar</i>	1.2546	(0.073)	1.4393	(0.095)
<i>gender · attr</i>	0.4567	(0.116)	0.4630	(0.153)
<i>gender · shar</i>	−0.1933	(0.103)	−0.2334	(0.134)
$\tau_{intercept}^2$	2.55		2.70	
τ_{gender}^2	−		0.00	
τ_{attr}^2	−		0.69	
τ_{shar}^2	−		0.49	
Covariate	B: random intercept		B: random slopes	
<i>intercept</i>	−0.6984	(0.121)	−0.8008	(0.128)
<i>gender</i>	0.1737	(0.165)	0.2737	(0.163)
<i>attr</i>	1.6370	(0.082)	1.9520	(0.116)
<i>shar</i>	1.2527	(0.076)	1.4466	(0.104)
<i>gender · attr</i>	0.4501	(0.118)	0.4603	(0.152)
<i>gender · shar</i>	−0.1877	(0.102)	−0.2310	(0.133)
$\tau_{intercept}^2$	2.61		2.77	
τ_{gender}^2	−		0.07	
τ_{attr}^2	−		0.76	
τ_{shar}^2	−		0.56	

measurements, at least to some extent. In addition, we also considered a model with random slopes for the main effects, i.e., the predictor is extended to

$$\tilde{\eta}_{ij} = \eta_{ij} + \gamma_{1i} \textit{gender}_i + \gamma_{2i} \textit{attr}_{ij} + \gamma_{3i} \textit{shar}_{ij},$$

and a diagonal matrix $\boldsymbol{Q} = \text{diag}(\tau_0^2, \tau_1^2, \tau_2^2, \tau_3^2)$ is assumed for $\boldsymbol{\gamma}_i \sim \mathcal{N}(\mathbf{0}, \boldsymbol{Q})$.

Table 7.4 contains estimates for the fixed effects and the random effects variances obtained through likelihood and Bayesian MCMC inference. Likelihood-based inference is carried out using function `lmer` of the R package `lme4`. Results for the Bayesian approach are obtained using `BayesX` with 102,000 iterations, 2,000 burn in iterations, and taking every 100th sampled parameter for inference. The hyperparameters for the inverse gamma variance parameters are $a = b = 0.001$. A sensitivity analysis with the alternatives $a = b = 0.01$ and $a = b = 0.0001$ shows that the results are relatively insensitive regarding the choice of hyperparameters. Only the variance parameter of the random slope for *gender* shows some sensitivity with a posterior mean ranging from 0.045 to 0.12.

As Table 7.4 reveals, the results for both approaches are in very close agreement. Attractiveness and shared interests have significant positive influence on the decision to see a partner again. Whereas the main effect of gender is not significant, the interactions of gender with the attributes reveal significant gender-specific differences. Attractiveness of the (female) partner increases the already positive effect on the decision of males to see her again, while shared interests are less important for males compared to females.

Table 7.5 Number of citations of patents: Poisson-normal model

Variable	Coefficient	Standard error	2.5 % quantile	97.5 % quantile
<i>intercept</i>	−0.347	0.061	−0.466	−0.227
<i>yearc</i>	−0.088	0.006	−0.100	−0.078
<i>ncountryc</i>	−0.036	0.008	−0.052	−0.019
<i>nclaimsc</i>	0.018	0.003	0.013	0.023
<i>biopharm</i>	0.174	0.061	0.051	0.290
<i>ustwin</i>	0.022	0.048	−0.074	0.112
<i>patus</i>	−0.163	0.051	−0.267	−0.063
<i>patgsgr</i>	−0.284	0.058	−0.400	−0.167
<i>opp</i>	0.399	0.045	0.316	0.489

The relatively large variance of the random intercept reflects the fact that there are still many individual-specific covariates or preferences not included in the model. Δ

Example 7.8 Number of Citations of Patents—Bayesian Linear Mixed Model

We analyze the Poisson-normal random intercept model of Example 7.6 (p. 392) using empirical and fully Bayesian inference. For fully Bayesian inference, we choose a noninformative flat prior for fixed effects and an inverse gamma distribution $\tau_0^2 \sim \text{IG}(a, b)$ with $a = b = 0.001$ for the variance parameter of the random intercepts. Results for empirical and fully Bayesian inference are similar. We therefore restrict the presentation to fully Bayesian inference which has been obtained using `bayesreg` objects of `BayesX`. The estimated fixed effects in Table 7.5, as well as standard errors, are in good agreement with the results in Example 5.7 (Table 5.10, p. 298). The estimated intercept −0.347 cannot be directly compared with the corresponding intercept 0.158 in Table 5.10, i.e., we have to adjust for $\tau_0^2/2$; see the remark on the Poisson-normal model in Sect. 7.5.2. After adjustment, the intercepts are in better agreement. The estimated value of τ_0^2 is $\hat{\tau}_0^2 = 1.17$ (posterior mean). Δ



7.7 Practical Application of Mixed Models

This section serves as a case study on (generalized) LMMs and how they can be applied in practice. Here, we restrict ourselves to models for longitudinal and cluster data.

7.7.1 General Guidelines and Recommendations

We first present some general guidelines and recommendations on how to apply LMMs or GLMMs in practice:

- *Exploratory analysis of cluster-specific heterogeneity:* Besides the usual inspection of the data through summary statistics and graphical tools (as outlined in Sect. 1.2) some additional graphical utilities may be useful for longitudinal and cluster data. An effective device is to perform a *cluster-wise analysis*.

In its simplest form, one could produce cluster-wise scatter plots of the relationship between the response and the continuous covariates as has been done in Example 2.9 (p. 39, Fig. 2.10) for the rats data. Such plots are also helpful to detect nonlinear relationships between the response and the covariates. Alternatively, one could estimate and visualize separate cluster-specific regression lines (or more generally regression curves), as illustrated in Sect. 7.1.1 (Figs. 7.1 and 7.4). For categorical covariates scatter plots can be replaced by box plots.

Additionally it is helpful to estimate the assumed regression model separately for each cluster (at least for the clusters where there are enough data for a separate analysis). The obtained cluster-specific intercepts and slopes could then be visualized in caterpillar plots together with 95 % confidence intervals. If there is cluster-specific heterogeneity, the cluster-specific intercepts and slopes should show considerable variability. We illustrate this approach in our case study below on sales of orange juice, as also seen in Fig. 7.10.

As mentioned, all cluster-wise analyses naturally require a sufficient amount of data in each cluster. If there are clusters with only a few observations, the cluster-wise analysis must be restricted to those clusters with sufficient data.

- *Model specification:* The specification of random intercepts and slopes is usually guided by subject-matter knowledge and a preceding exploratory analysis. When specifying random coefficient models, one should always think of possibly different between- and within-cluster effects or endogenous (stochastic) covariates; recall the discussions in Sect. 7.1.1 (p. 353) and in Sect. 7.1.5 (p. 366).

Usually it does not make sense to include random slopes without including a random intercept. It is also typically not sensible to include a random slope without including a corresponding fixed effect. Generally, the specification of random slopes should be as parsimonious as possible. Often convergence problems of estimation algorithms are encountered if too complex models are specified. A nice discussion of the pitfalls in the specification of random coefficient models is given in Sect. 4.10 of [Skrondal and Rabe-Hesketh \(2008\)](#).

- *Model assessment and diagnostics:* In addition to the usual model diagnostic tools discussed in Sect. 3.4.4 (p. 155), tools similar to those used for the exploratory analysis are helpful for model assessment in random coefficients models. In particular, caterpillar plots of the estimated random coefficients help to assess the heterogeneity assumption. It may also be useful to visualize the estimated cluster-specific curves together with partial residuals and kernel densities of random effects to assess the normality assumption (at least to a certain extent). The results of hypothesis tests could be used as an additional source of information for model diagnostics, but should be taken with care; see the discussion in Sect. 7.3.4.

Since there are a considerable number of competing estimation approaches, it is recommended to estimate the models using these different approaches (ML, REML, fully Bayesian under different choices of hyperparameters) and even different software to assess the stability of estimates and to investigate possible convergence problems (at least when using complex model specifications).

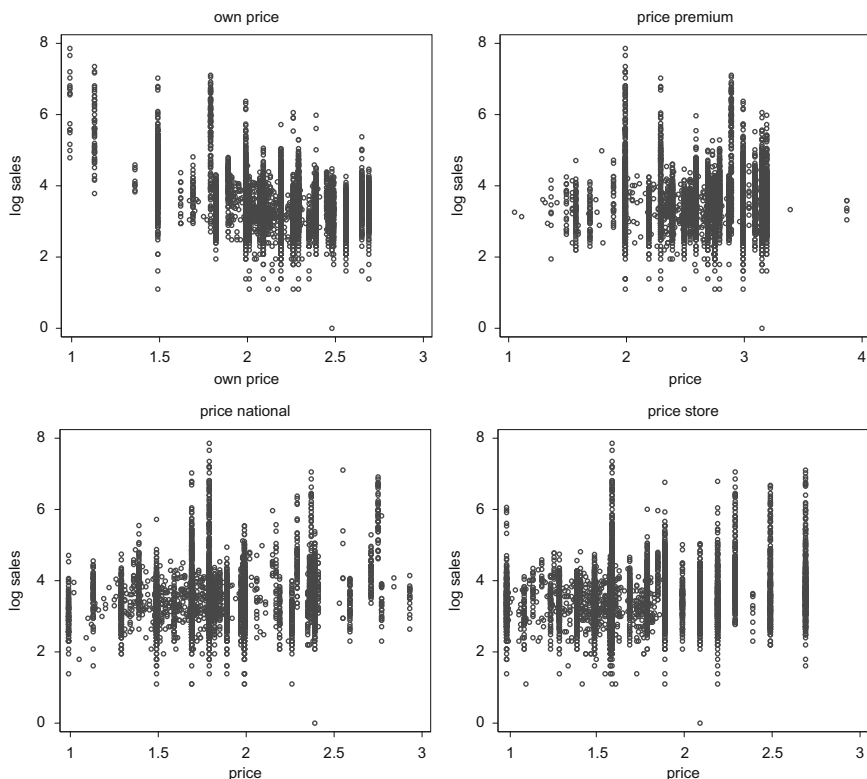


Fig. 7.8 Sales of orange juice: scatter plots between the log number of packages sold and the price of “tree fresh” (*upper left panel*), as well as prices of competitive brands

7.7.2 Case Study on Sales of Orange Juice

Data Description and Exploratory Analysis

We now illustrate the application of LMMs using the supermarket scanner data that were briefly outlined in Example 4.9 (p. 199) in the context of multiplicative errors. The data include weekly unit sales and respective retail prices for different brands of orange juice (premium, national, and store brands) in 81 chains of a store over a time span of 89 weeks. We restrict ourselves to one of the national brands, i.e., the “Tree Fresh” brand. In the following, we model the log sales y_{ij} in store i , $i = 1, \dots, 81$, at time t_{ij} in relation to its own-item price p_{ij} and competitor prices $pc1_{ij}$, $pc2_{ij}$, and $pc3_{ij}$ among the three quality tiers (premium, national, store). To get an intuitive feel for the data, Fig. 7.8 shows scatter plots of log sales against the covariates. We clearly find an inverse relationship between unit sales and



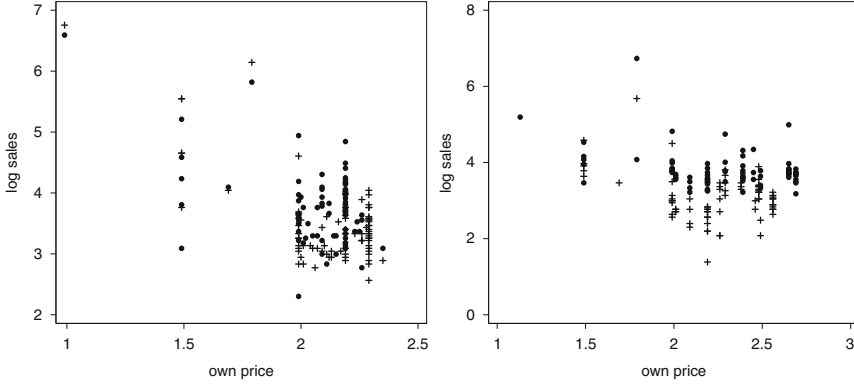


Fig. 7.9 Sales of orange juice: scatter plots between the log number of sold packages and the own price in four different stores. Each plot displays the log sales versus price in two outlets using different symbols (*plus signs and black dots*)

own-item price. The relationship also appears to be nonlinear as the slope decreases for prices higher than values near 2. However, the situation is less clear for the impact of competitive prices on the sales of Tree Fresh, i.e., it is much more difficult to discern any relationships in these scatter plots. Of course, economic theory suggests a positive correlation between sales and prices of competitors, which can be observed at least for the price of the store brand. In order to investigate possible store-specific heterogeneity, it is instructive to plot the log sales price relationship for different stores. Figure 7.9 displays log sales against own-item price for four different stores exhibiting a clear tendency for outlet-specific heterogeneity. It is apparent that the overall sales level and also the slope of the relationship vary across the four stores. The scatter plots further suggest transforming prices to log prices in order to satisfactorily capture the nonlinearity in the relationships. To further check cluster-specific heterogeneity, we estimate the model

$$y_{ij} = \beta_{0i} + \beta_{1i} \cdot \log p_{ij} + \beta_{2i} \cdot \log pc1_{ij} + \beta_{3i} \cdot \log pc2_{ij} + \beta_{4i} \cdot \log pc3_{ij} + \varepsilon_{ij} \quad (7.47)$$

separately for each store $i = 1, \dots, 81$. In this model, $\beta_{1i}, \dots, \beta_{4i}$ are fixed effects of the log price and the log prices of competitors. All covariates are centered about their mean values to avoid problems with meaningless origins. Figure 7.10 shows caterpillar plots of the estimated cluster-specific coefficients. We clearly see that the intercepts, as well as the regression coefficients, of the own-item price shows considerable cluster-specific heterogeneity. The regression coefficients of the cross-price effects reveal much less heterogeneity.

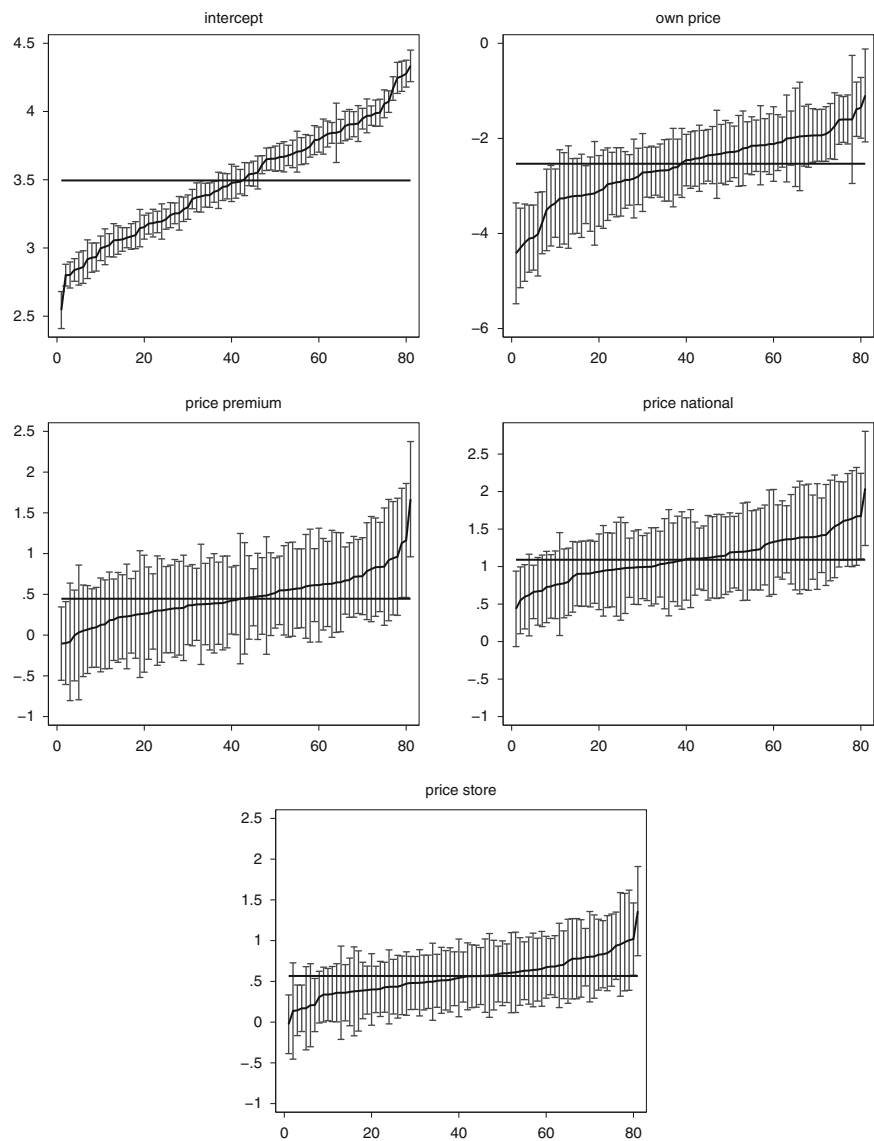


Fig. 7.10 Sales of orange juice: caterpillar plots of the regression coefficients in a cluster-wise analysis of the regression model (7.47). Shown are the estimated regression coefficients and their 95 % confidence intervals. The solid lines correspond to the regression coefficients of a global regression model

Table 7.6 Sales of orange juice: estimated fixed effects and variance parameters for the full model (7.48) and a reduced model without random slopes for cross-price effects

		Full		Reduced	
	Parameter	Estimate	s.e.	Estimate	s.e.
<i>intercept</i>	β_0	3.489	0.045	3.490	0.045
<i>logp</i>	β_1	−2.558	0.077	−2.533	0.074
<i>logpc1</i>	β_2	0.454	0.036	0.453	0.033
<i>logpc2</i>	β_3	1.114	0.038	1.107	0.031
<i>logpc3</i>	β_3	0.549	0.029	0.535	0.023
$\text{Var}(\gamma_{0i})$	τ_0^2	0.166		0.165	
$\text{Var}(\gamma_{1i})$	τ_1^2	0.354		0.309	
$\text{Var}(\gamma_{2i})$	τ_2^2	0.020		—	
$\text{Var}(\gamma_{3i})$	τ_3^2	0.041		—	
$\text{Var}(\gamma_{4i})$	τ_4^2	0.023		—	
$\text{Var}(\varepsilon_{ij})$	σ^2	0.186		1.190	

Model Specification and Results

In light of our discussion and exploratory analysis, we estimate the two-level random coefficients model

$$y_{ij} = (\beta_0 + \gamma_{0i}) + (\beta_1 + \gamma_{1i}) \cdot \log p_{ij} + (\beta_2 + \gamma_{2i}) \cdot \log pc1_{ij} + (\beta_3 + \gamma_{3i}) \cdot \log pc2_{ij} + (\beta_4 + \gamma_{4i}) \cdot \log pc3_{ij} + \varepsilon_{ij}. \tag{7.48}$$

Here, $\beta_0 + \gamma_{0i}$ are store-specific random intercepts, and $\gamma_{1i}, \dots, \gamma_{4i}$ are store-specific deviations from the overall slopes of price effects. For the random effects, a fully unspecified covariance matrix is assumed a priori. Estimation has been carried out using function `lmer` of the R package `lme4`.

Table 7.6 shows the estimated fixed effects and variance parameters. The estimated log-linear price effects are in agreement with economic theory. Increasing own prices reduce sales, while increasing prices of competing brands lead to increased sales for “Tree Fresh.” The strength of the outlet-specific random variation differs across stores, measured through the random effects variances. While the random intercept (measuring the outlet-specific overall sales level) and the own-price random effects are comparably strong with variances of $\tau_0^2 = 0.166$ and $\tau_1^2 = 0.354$ (in relation to the overall variance of $\sigma^2 = 0.186$), the random effects of cross-price effects are comparably small. Note that a comparison of random effects variances only makes sense if the covariates are measured on the same scale, which is the case for all price covariates.

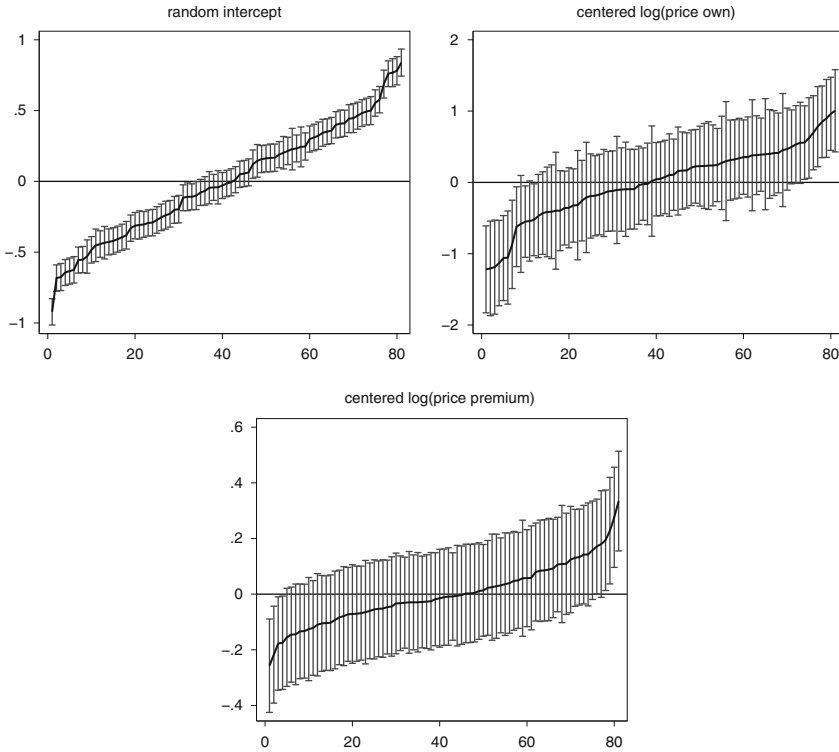


Fig. 7.11 Sales of orange juice: caterpillar plots of some random effects in the full model (7.48). The estimated regression coefficients and their 95 % confidence intervals are shown

Model Diagnostics

Figure 7.11 displays caterpillar plots of the estimated random effects, while Fig. 7.12 shows kernel densities of the random effects. Both figures confirm that the random effects of cross-prices are small compared to the random intercept and the random slope of the own-item price. Another way to get an intuition about the store-specific heterogeneity is to plot the store-specific price curves. This is done for four outlets in Fig. 7.13 showing considerable own-price heterogeneity.

To test the stability of the results, model (7.48) has been reestimated using a fully Bayesian approach based on MCMC methods and the software package `BayesX`. The estimates are built on 102,000 iterations, with 2,000 iterations as burn in period and every 100th sampled parameter used for subsequent inference. We assumed independent inverse gamma priors for the variance parameters $\tau_0^2, \dots, \tau_4^2$ with hyperparameters $a = b = 0.001$. The results remain stable for the alternative choices $a = b = 0.01$ and $a = b = 0.0001$. Table 7.7 shows the resulting fixed effects and random effects variance parameter estimates. Overall the results are in close agreement with the results of `lme4` given in Table 7.6.

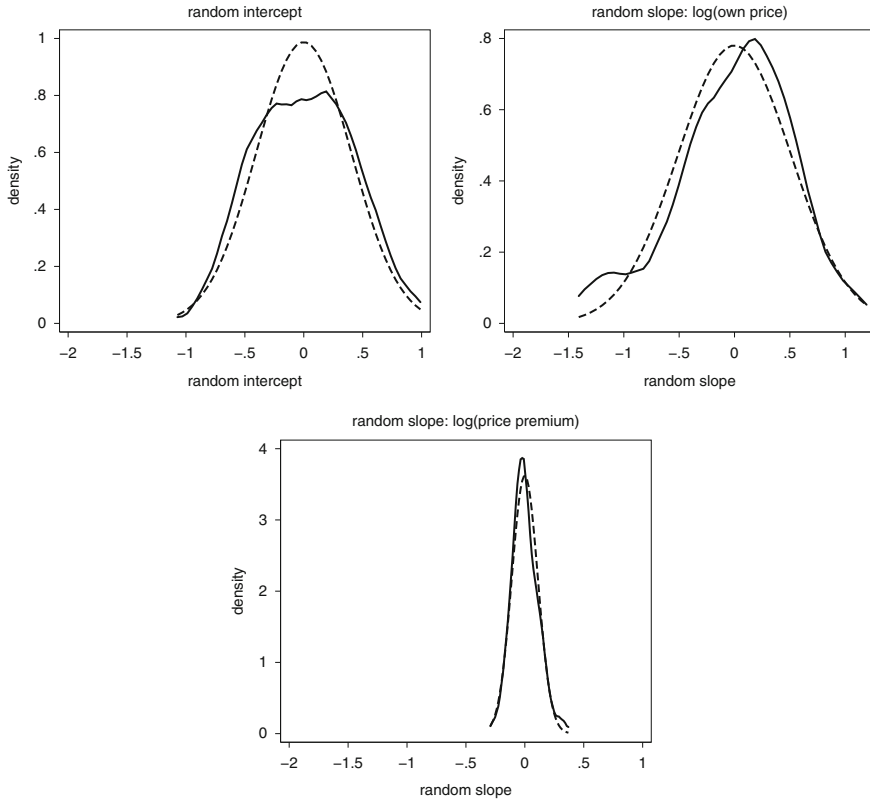


Fig. 7.12 Sales of orange juice: kernel densities of random effects (*solid lines*) together with corresponding normal densities (*dashed lines*) in the full model (7.48)

Since the estimated variances of cross-price random slopes are small, it seems reasonable to simplify the model by removing all or some of the cross-price random effects. Indeed, if we apply the simulation-based restricted likelihood ratio test (function `exactRLRT` of the R package `RLRsim`) for testing $H_0 : \tau_l^2 = 0$, $l = 2, 3, 4$ versus $H_1 : \tau_l^2 \neq 0$, most p-values are comparably large indicating that the null hypotheses cannot be rejected. The only p-value lower than 0.05 is for *logpc3*. Moreover, the DIC in a fully Bayesian approach for a reduced model without cross-price random effects is only 11 points above that of the full model (7.48). Hence, these results are at least supportive that a reduced model without cross-price random slopes may be sufficient for these data. The results for the reduced model are again given in Tables 7.6 and 7.7.

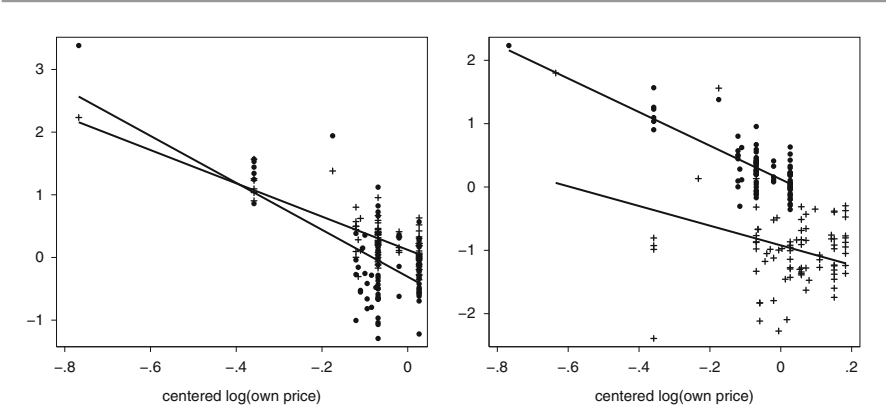


Fig. 7.13 Sales of orange juice: log sales versus log price curves for four different outlets together with corresponding partial residuals in the full model (7.48)

Table 7.7 Sales of orange juice: estimated fixed effects and variance parameters for the full model (7.48) and a reduced model without random slopes for cross-price effects in a fully Bayesian approach

		Full		Reduced	
	Parameter	Estimate	s.e.	Estimate	s.e.
<i>intercept</i>	β_0	3.489	0.046	3.488	0.047
<i>logp</i>	β_1	-2.540	0.076	-2.532	0.074
<i>logpc1</i>	β_2	0.454	0.033	0.453	0.033
<i>logpc2</i>	β_3	1.110	0.034	1.108	0.031
<i>logpc3</i>	β_4	0.542	0.028	0.535	0.023
$\text{Var}(\gamma_{0i})$	τ_0^2	0.171		0.169	
$\text{Var}(\gamma_{1i})$	τ_1^2	0.320		0.317	
$\text{Var}(\gamma_{2i})$	τ_2^2	0.008		—	
$\text{Var}(\gamma_{3i})$	τ_3^2	0.014		—	
$\text{Var}(\gamma_{4i})$	τ_4^2	0.017		—	
$\text{Var}(\varepsilon_{ij})$	σ^2	0.188		1.190	

7.8 Bibliographic Notes and Proofs

7.8.1 Bibliographic Notes

LMMs for longitudinal data are described extensively in [Verbeke and Molenberghs \(2000\)](#), with a focus on likelihood inference and biostatistical applications. Other books including (chapters on) LMMs and GLMMs are [Diggle et al. \(2002\)](#) (also with biostatistical applications), [Demidenko \(2004\)](#), [Skrondal and Rabe-Hesketh \(2004\)](#), and the handbook on longitudinal data edited by [Fitzmaurice, Davidian, Verbeke, and Molenberghs \(2003\)](#). [Gelman and Hill \(2006\)](#) place more emphasis on

clustered and, more generally, multilevel data and on social science applications. LMMs from an econometricians point of view are covered in [Hsiao \(2003\)](#). Bayesian LMM and GLMM are described in [Fahrmeir and Kneib \(2011\)](#), Chap. 3, including models with non-Gaussian random effects priors. A nice introduction to LMM for longitudinal and cluster data from a practitioner's point of view is given in [Skrondal and Rabe-Hesketh \(2008\)](#). Although the focus is on analysis with STATA, it is worth reading even if STATA is not used for inference.

Besides mixed models, two other conceptual models exist for the analysis of longitudinal data. *Conditional* or *autoregressive models* extend linear and generalized linear models for cross-sectional data by including effects of response values observed at previous time points $j - 1, j - 2, \dots$ into the predictor. The simplest possibility is

$$\eta_{ij} = \mathbf{x}'_{ij}\boldsymbol{\beta} + y_{i,j-1}\alpha,$$

i.e., an autoregressive predictor of first order. For statistical inference, $y_{i,j-1}$ is treated formally as an additional covariate.

Marginal models, often in form of generalized estimating equations (*GEE*) approaches, consider multivariate versions of score functions with similar structure as the score functions in Sect. 6.4 for categorical response. However, the true covariance matrix $\boldsymbol{\Sigma}_i$ of the multinomial distribution is replaced through a “working covariance matrix” $\mathbf{S}_i$, leading to a multivariate version of the quasi-likelihood approach in Sect. 5.5. Solving the estimating equation $s(\hat{\boldsymbol{\beta}}) = \mathbf{0}$ provides the quasi-likelihood or GEE estimator $\hat{\boldsymbol{\beta}}$ in marginal models.

Both classes of models are described, for example, in [Diggle et al. \(2002\)](#) and, in less detail, in [Fahrmeir and Tutz \(2001\)](#).

7.8.2 Proofs

Estimator in the Fixed Effects Model (P. 367)

We first consider the classical linear model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$$

involving two sets of covariates with design matrices $\mathbf{X}_1$ and $\mathbf{X}_2$. We derive the least squares estimator for $\boldsymbol{\beta}_2$. The normal equations are given by

$$\begin{pmatrix} \mathbf{X}'_1\mathbf{X}_1 & \mathbf{X}'_1\mathbf{X}_2 \\ \mathbf{X}'_2\mathbf{X}_1 & \mathbf{X}'_2\mathbf{X}_2 \end{pmatrix} \begin{pmatrix} \boldsymbol{\beta}_1 \\ \boldsymbol{\beta}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{X}'_1\mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}'_1\mathbf{X}_2\boldsymbol{\beta}_2 \\ \mathbf{X}'_2\mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}'_2\mathbf{X}_2\boldsymbol{\beta}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{X}'_1\mathbf{y} \\ \mathbf{X}'_2\mathbf{y} \end{pmatrix}.$$

Solving the first set of equations for $\boldsymbol{\beta}_1$ gives the solution

$$\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}'_1\mathbf{X}_1)^{-1}\mathbf{X}'_1\mathbf{y} - (\mathbf{X}'_1\mathbf{X}_1)^{-1}\mathbf{X}'_1\mathbf{X}_2\hat{\boldsymbol{\beta}}_2 = (\mathbf{X}'_1\mathbf{X}_1)^{-1}\mathbf{X}'_1(\mathbf{y} - \mathbf{X}_2\hat{\boldsymbol{\beta}}_2). \quad (7.49)$$

Thus, the estimator for β_1 is obtained by regressing the covariates in X_1 on the partial residuals $y - X_2\hat{\beta}_2$. Inserting Eq. (7.49) in the second equation of the normal equations yields

$$X_2'X_1(X_1'X_1)^{-1}X_1'y - X_2'X_1(X_1'X_1)^{-1}X_1'X_2\hat{\beta}_2 + X_2'X_2\hat{\beta}_2 = X_2'y.$$

Rearranging terms then gives

$$X_2'X_2\hat{\beta}_2 - X_2'X_1(X_1'X_1)^{-1}X_1'X_2\hat{\beta}_2 = X_2'(I - X_1(X_1'X_1)^{-1}X_1')y.$$

Solving for $\hat{\beta}_2$ finally produces

$$\begin{aligned}\hat{\beta}_2 &= (X_2'(I - X_1(X_1'X_1)^{-1}X_1')X_2)^{-1} (X_2'(I - X_1(X_1'X_1)^{-1}X_1')y) \\ &= (X_2'(I - H_1)X_2)^{-1} (X_2'(I - H_1)y),\end{aligned}$$

where H_1 is the hat matrix (see Box 3.5 on p. 108) of a regression with the variables in X_1 as the regressors. Note that $(I - H_1)X_2$ is a matrix of residuals in regressions of the variables in X_1 on the variables in X_2 . Since both H_1 and $I - H_1$ are idempotent, we obtain

$$\hat{\beta}_2 = (\tilde{X}_2'\tilde{X}_2)^{-1} \tilde{X}_2'\tilde{y},$$

where $\tilde{X}_2 = (I - H_1)X_2$ and $\tilde{y} = (I - H_1)y$. Thus, the estimator $\hat{\beta}_2$ is obtained in two steps. We first regress each column of X_2 on the variables in X_1 . In a second step the set of residuals from the first regression is regressed on the residuals of a regression of X_1 on y . While this is an important result on its own, we use it to derive the estimator for the regression coefficients in the fixed effects model of p. 367. Consider the model

$$y = X\beta + Z\gamma_0 + \varepsilon,$$

where X is the design matrix of the covariates (excluding the intercept) and $Z = (z_1, \dots, z_m)$ with z_i being a dummy vector indicating the i th unit or cluster. Applying our result on partitioned regression, we obtain

$$\hat{\beta} = (X'(I - H_Z)X)^{-1}X'((I - H_Z)y),$$

with $H_Z = Z(Z'Z)^{-1}Z'$. This corresponds to a least squares regression using the transformed design matrix $(I - H_Z)X$ and the transformed response vector $(I - H_Z)y$. Due to the specific structure of Z , it is easily verified that premultiplication of $(I - H_Z)$ results in cluster-wise centering of the columns of X and of y . Thus we obtain $\hat{\beta}$ using a regression of the cluster-wise centered covariates $x_{ijl} - \bar{x}_{il}$ on the cluster-wise centered responses $y_{ij} - \bar{y}_i$.

Derivation of the Penalized Least Squares Estimator in the LMM (Sect. 7.3.1 on p. 371)

We first rearrange the penalized least squares criterion to obtain

$$\begin{aligned}
 \text{LS}_{\text{pen}}(\beta, \gamma) &= (y - X\beta - U\gamma)'R^{-1}(y - X\beta - U\gamma) + \gamma'G^{-1}\gamma \\
 &= y'R^{-1}y - y'R^{-1}X\beta - y'R^{-1}U\gamma \\
 &\quad - \beta'X'R^{-1}y + \beta'X'R^{-1}X\beta + \beta'X'R^{-1}U\gamma \\
 &\quad - \gamma'U'R^{-1}y + \gamma'U'R^{-1}X\beta + \gamma'U'R^{-1}U\gamma \\
 &\quad + \gamma'G^{-1}\gamma \\
 &= y'R^{-1}y - 2y'R^{-1}X\beta + \beta'X'R^{-1}X\beta + 2\gamma'U'R^{-1}X\beta \\
 &\quad - 2y'R^{-1}U\gamma + \gamma'(U'R^{-1}U + G^{-1})\gamma.
 \end{aligned}$$

Applying two rules for the differentiation of vector functions [see Theorem A.33 (1) and (3)], we obtain the derivatives

$$\begin{aligned}
 \frac{\partial \text{LS}_{\text{pen}}(\beta, \gamma)}{\partial \beta} &= -2X'R^{-1}y + 2X'R^{-1}X\beta + 2X'R^{-1}U\gamma, \\
 \frac{\partial \text{LS}_{\text{pen}}(\beta, \gamma)}{\partial \gamma} &= -2U'R^{-1}y + 2(U'R^{-1}U + G^{-1})\gamma + 2U'R^{-1}X\beta.
 \end{aligned}$$

This results in the equation system

$$\begin{pmatrix} X'R^{-1}X & X'R^{-1}U \\ U'R^{-1}X & U'R^{-1}U + G^{-1} \end{pmatrix} \begin{pmatrix} \hat{\beta} \\ \hat{\gamma} \end{pmatrix} = \begin{pmatrix} X'R^{-1}y \\ U'R^{-1}y \end{pmatrix}.$$

Defining $C = (X, U)$ and the partitioned matrix

$$B = \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & G^{-1} \end{pmatrix},$$

yields

$$(C'R^{-1}C + B) \begin{pmatrix} \beta \\ \gamma \end{pmatrix} = C'R^{-1}y.$$

The solution is the penalized least squares estimator (7.25).